

Practice Questions

LEVEL I

(Questions based on Fundamentals)

ABC OF BINOMIAL EXPANSION

- Write the expansion of $(x+1)^6$.
- Write the expansion of $\left(\frac{x}{5} + \frac{1}{x}\right)^5$.
- Find the degree of the polynomial $(x + \sqrt{x^2-1})^6 + (x - \sqrt{x^2-1})^6$.
- Find the number of terms in the expansion of
 - $(1+x)^{2013}$
 - $(1+2x+x^2)^{1007}$
 - $(1+3x+3x^2+x^3)^{668}$
 - $(1+4x+6x^2+4x^3+x^4)^{503}$
 - $(1+x)(1+x^2)^{10}$
 - $(1-x)^4(1+x+x^2)^{10}$
- Find the number of terms in the expansion of $(a+b)^{50} + (a-b)^{50}$.
- Find the number of terms in the expansion of $\left(x + \frac{1}{x}\right)^{99} + \left(x - \frac{1}{x}\right)^{99}$.
- Find the number of terms in the expansion of $\left(x^2 + \frac{1}{x^2}\right)^{20} - \left(x^2 - \frac{1}{x^2}\right)^{20}$.
- Find the number of terms in the expansion of $\left(\frac{2}{x} + x^3\right)^{2015} - \left(\frac{2}{x} - x^3\right)^{2015}$.

GENERAL TERM IN A BINOMIAL EXPANSION

- Find the 13th term in the expansion of $\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}$.
- Find the 4th term from the end in the expansion of $\left(\frac{3}{x^2} - \frac{x^3}{6}\right)^7$.
- Find the term independent of x in the expansion of $\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^6$.
- Find the co-efficient of x^7 in the expansion of $\left(3x^2 + \frac{1}{5x}\right)^{11}$.
- Determine the value of x in the expansion of $(x + x^{\log_{10} x})^5$, where the 3rd term is 10,000.
- Prove that the co-efficient of x^n in $(1+x)^{2n}$ is twice the co-efficient of x^n in $(1+x)^{2n-1}$.

- If the co-efficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^{2n}$ are in AP, prove that $2n^2 - 9n + 7 = 0$.
- In the binomial expansion of $(1+y)^n$, where n is a natural number, the co-efficients of the 5th, 6th and 7th terms are in AP. Find the value of n .
- Let n be a positive integer. If the co-efficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^n$ are in AP, find n .
- Find the co-efficient of x^{10} in $(1+x)^{50}$.
- Find the co-efficient of x^{10} in $(1+x^2)^{30}$.
- Find the co-efficient of x^{50} in the expansion of $(1+x)^{41}(1-x+x^2)^{40}$.
- Find the co-efficients of x, x^2, x^3 in the expansion of $(1+x)^5 \times (1-x)^6$.
- If in the expansion of $(1+x)^m(1-x)^n$, the co-efficient of x and x^2 are 3 and -6, respectively, find m and n .
- Find the middle term in the expansion of $\left(\frac{2x^2}{3} - \frac{3}{2x}\right)^{12}$.
- Find the middle term in the expansion of $\left(3x - \frac{x^3}{6}\right)^9$.
- Prove that the middle term in the expansion of $(1+x)^{2n}$ is $\frac{1.2.3 \dots (2n-1)}{(n)!} \times 2^n \times x^n$.
- Prove that the co-efficients of the middle terms $(1+x)^{2n}$ is equal to the sum of the co-efficients of the two middle terms in $(1+x)^{2n-1}$.

GREATEST CO-EFFICIENTS/TERMS OF A BINOMIAL EXP

- Find the greatest co-efficient of the polynomial $\left(\frac{1}{4} + \frac{2}{3}x\right)^{100}$.
- If the sum of the co-efficients in the expansion of $(a+b)^n$ is 4096, then find the greatest co-efficient in the given expansion.
- If the sum of the co-efficients in the expansion of $(1+2x)^n$ is 6561, find the greatest co-efficient in the given expansion.
- Find the greatest term in the expansion of $(2+3x)^{10}$, when $x = 3/5$.
- Find the greatest term in the expansion of $(3-5x)^{15}$, when $x = 1/5$.

DIVISIBILITY RELATED PROBLEMS

- If n be a positive integer, prove that $(4^n - 3n - 1)$ is divisible by 9.
- If n be a positive integer, prove that $3^{2n+2} - 8n - 9$ is divisible by 64.
- Prove that $11^{n+2} + 12^{2n+1}$ is divisible by 133.

35. Prove that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for all n .
 36. Prove that $11^{10} - 1$ is divisible by 100.
 37. Find the remainder when 7^{98} is divided by 5.
 38. Prove that $1^{2013} + 2^{2013} + 3^{2013} + \dots + 2011^{2013} + 2012^{2013}$ is divisible by 2013.
 39. Prove that $1992^{1998} - 1955^{1998} - 1938^{1998} + 1901^{1998}$ is divisible by 1998.
 40. Prove that $53^{53} - 33^3$ is divisible by 10.

UNIT DIGIT OF A NATURAL NUMBER

41. Find the unit digit of $(27)^{50} + (18)^{50}$.
 42. Find the unit digit of $1! + 2! + 3! + 4! + 5! + \dots + (33)!$.
 43. Find the last two digits of $(27)^{27}$.
 44. Find the last two digits of 3^{999} .
 45. Find the last two digit of $(17)^{10}$.

RATIONAL TERM IN A BINOMIAL EXPANSION

46. Find the number of integral terms in the expansion of $(9^{1/4} + 8^{1/6})^{500}$.
 47. Find the number of irrational terms in the expansion of $(3^{1/5} + 2^{1/3})^{1000}$.
 48. Find the number of non-integral terms in the expansion of $(5^{1/8} + 2^{1/6})^{100}$.
 49. Find the sum of rational terms in the expansion of $(2^{1/2} + 3^{1/5})^{10}$.

INTEGRAL AND FRACTIONAL PART OF A NUMBER

50. If $(\sqrt{P} + Q)^n = I + f$, where I and n are natural numbers, n being odd and $0 \leq f < 1$, show that $(I + f)f = k^n$, where $P - Q^2 = k$ and $\sqrt{Q} - P < 1$.
 51. If $(\sqrt{P} + Q)^n = I + f$, where I and n are natural numbers, n being even and $0 \leq f < 1$, show that $(I + f)(I - f)f = k^n$, where $P - Q^2 = k$ and $\sqrt{Q} - P < 1$.
 52. Prove that the integral part of $(5\sqrt{5} + 11)^{2n+1}$ is even when $n \in N$ and hence if $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$, where $[\] = \text{GIF}$, prove that $Rf = 4^{2n+1}$.
 53. If $x = (8 + 3\sqrt{7})^n$, $n \in N$, prove that the integral part of x is an odd integer and show that $x - x^2 + x[x] = 1$, where $[\] = \text{GIF}$.
 54. If $(8 + 3\sqrt{7})^n = \alpha + \beta$, where n and α are positive integers and β is a positive proper fraction, prove that $(1 - \beta)(\alpha + \beta) = 1$.
 55. Find the integer just greater than the number $(3 + \sqrt{5})^5$.

PROPERTIES OF BINOMIAL CO-EFFICIENTS

56. Prove that the co-efficients of a^n and a^m in the expansion of $(1 + a)^{m+n}$ are equal.
 57. If $^{10}C_r = ^{10}C_{r+4}$, find the value of r .
 58. Prove that

$$\frac{C_1}{C_0} + 2 \cdot \frac{C_2}{C_1} + 3 \cdot \frac{C_3}{C_2} + \dots + n \cdot \frac{C_n}{C_{n-1}} = \frac{n(n+1)}{2}.$$

59. Find the value of

$$\frac{^{15}C_1}{^{15}C_0} + 2 \cdot \frac{^{15}C_2}{^{15}C_1} + 3 \cdot \frac{^{15}C_3}{^{15}C_2} + \dots + 15 \cdot \frac{^{15}C_{15}}{^{15}C_{14}}.$$

60. Find the value of

$$\left(1 + \frac{C_1}{C_0}\right) \left(1 + \frac{C_2}{C_1}\right) \left(1 + \frac{C_3}{C_2}\right) \dots \left(1 + \frac{C_n}{C_{n-1}}\right)$$

61. Find the value of

$$(^nC_0 + ^nC_1)(^nC_1 + ^nC_2)(^nC_2 + ^nC_3) \dots (^nC_n + ^nC_{n-1}),$$

if $^nC_0 \cdot ^nC_1 \cdot ^nC_2 \dots ^nC_{n-1} = k$

62. Prove that $^nC_r + ^nC_{r-1} = ^{n+1}C_r$.

63. Evaluate $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}$.

64. Evaluate: $\binom{n}{r} + 3\binom{n}{r-1} + 3\binom{n}{r-2} + \binom{n}{r-3}$.

65. Evaluate:

$$\binom{n}{r} + 4\binom{n}{r-1} + 6\binom{n}{r-2} + 4\binom{n}{r-3} + \binom{n}{r-4}.$$

66. Evaluate: $^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3$.

67. Prove that

$$^mC_m + ^{m+1}C_m + ^{m+2}C_m + \dots + ^nC_m = ^{n+1}C_{m+1}.$$

68. Find the value of

$$^{20}C_{20} + ^{21}C_{20} + ^{22}C_{20} + ^{23}C_{20} + \dots + ^{2014}C_{20}.$$

69. Find the value of

$$^{48}C_{47} + ^{49}C_{47} + ^{50}C_{47} + ^{51}C_{47} + \dots + ^{1005}C_{47}.$$

70. Prove that

$$^nC_r + ^{n-1}C_r + ^{n-2}C_r + ^{n-3}C_r + \dots + ^rC_r = ^{n+1}C_{r+1}.$$

71. Find the value of

$$^{49}C_{10} + ^{48}C_{10} + ^{47}C_{10} + \dots + ^{10}C_{10}.$$

72. Find the value of

$$^{2014}C_{100} + ^{2013}C_{100} + ^{2012}C_{100} + \dots + ^{100}C_{100}.$$

73. Find the value of

$$^{70}C_{20} + ^{69}C_{20} + ^{68}C_{20} + ^{67}C_{20} + \dots + ^{21}C_{20}.$$

74. Prove that

$$^{2n}C_0 + ^{2n}C_1 + ^{2n}C_2 + ^{2n}C_3 + \dots + ^{2n}C_n = 2^{2n-1} + \frac{1}{2} {}^{2n}C_n.$$

75. Find the sum of $^{30}C_0 + ^{30}C_1 + ^{30}C_2 + \dots + ^{30}C_{15}$.

76. Find the sum of $^{40}C_{20} + ^{40}C_{21} + ^{40}C_{22} + \dots + ^{40}C_{40}$.

77. Prove that

$$^nC_3 + 2 \cdot ^{n+1}C_3 + 3 \cdot ^{n+2}C_3 + \dots + n \cdot ^{2n-2}C_3 = -^{2n}C_5 + ^nC_5 + n \cdot ^{2n}C_4.$$

78. Find the sum of

$$^{100}C_{50} + 2 \cdot ^{99}C_{49} + 3 \cdot ^{98}C_{48} + \dots + 51 \cdot ^{50}C_0$$

79. Find the co-efficient of x^{50} in the expansion of $(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000}$.

80. Find the sum of $C_1 + 2 \cdot C_2 + 3 \cdot C_3 + \dots + n \cdot C_n$.
 81. Find the sum of $C_0 + 2 \cdot C_1 + 3 \cdot C_2 + \dots + (n+1) \cdot C_n$.
 82. Find the sum of $2 \cdot C_1 + 2^2 \cdot C_2 + 3^2 \cdot C_3 + \dots + n^2 \cdot C_n$.
 83. Find the sum of

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1}.$$

84. Find the sum of

$$\frac{C_0}{1.2} + \frac{C_1}{2.3} + \frac{C_2}{3.4} + \dots + \frac{C_n}{(n+1)(n+2)}.$$

85. Prove that

$${}^nC_r \cdot {}^nC_0 + {}^nC_{r-1} \cdot {}^nC_1 + {}^nC_{r-2} \cdot {}^nC_2 + \dots + {}^nC_m \cdot {}^nC_r = {}^{m+n}C_r$$

86. Find the sum of

$$C_0 \cdot C_n + C_1 \cdot C_{n-1} + C_2 \cdot C_{n-2} + \dots + C_n \cdot C_0.$$

87. Prove that

$${}^nC_r \cdot {}^nC_n + {}^nC_{r-1} \cdot {}^nC_{n-1} + {}^nC_{r-2} \cdot {}^nC_{n-2} + \dots + {}^nC_0 \cdot {}^nC_{n-r} = 2^n C_{n-r}$$

88. Find the sum of

$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = 2^n C_n.$$

MULTINOMIAL THEOREM

89. Find the sum of

$$C_0 \cdot C_1 + C_1 \cdot C_2 + C_2 \cdot C_3 + \dots + C_{n-1} \cdot C_n.$$

90. Find the co-efficient of $a^3b^4c^2$ in the expansion of $(a+b+c+d)^{15}$.

91. Find the co-efficient of $a^3b^4c^7$ in the expansion of $(ab+bc+ca)^8$.

92. Find the greatest co-efficient of $(a+b+c+d+e)^{53}$.

93. Find the number of terms in the expansion of

- (i) $(a+b+c)^n$
 (ii) $(a+b+c+d)^n$
 (iii) $(a+b+c+d+e)^n$.

94. Find the co-efficient of x^7 in the expansion of $(1+3x-2x^3)^{10}$.

BINOMIAL THEOREM FOR ANY INDEX

95. Find the co-efficient of x^n in the expansion of $\frac{1}{1-3x}$.

96. Find the co-efficient of x^n in the expansion of $\frac{1}{1+4x}$.

97. Find the co-efficient of x^n in the expansion of $\frac{1}{1-9x+20x^2}$.

98. Find the co-efficient of x^n in the expansion of $\frac{1}{1-(a+b)x+abx^2}$.

99. Find the co-efficient of x^n in the expansion of $\left(\frac{1+x}{1-x}\right)$.

100. Find the co-efficient of x^n in the expansion of $\left(\frac{1+x}{1-x}\right)^2$.

101. Find the co-efficient of x^n in the expansion of $(1-2x)^{-1/2}$.

102. Find the first negative term in the expansion of $\left(1+\frac{3}{4}x\right)^{13/3}$, where $0 < x < 4/3$.

103. If $y = x - x^2 + x^3 - x^4 + \dots$, where $|x| < 1$, prove that $x = \frac{y}{1-y}$.

104. If $y = 2x + 3x^2 + 4x^3 + \dots$, where $|x| < 1$, prove that $x = 1 - \frac{1}{\sqrt{1+y}}$.

105. Find the sum of the series

$$1 + \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$$

106. If x be so small that its square and higher powers may be neglected, find the value of

$$\frac{(1+2x)^{1/2}(16+3x)^{1/4}}{(1-x)^2} \text{ at } x = \frac{1}{2}.$$

107. If for small values of x , $\frac{(1+2x)^{1/2} + (16+3x)^{1/4}}{(1-2x)^{1/4}}$ is very

nearly equal to $a + bx$, find the values of a and b .

108. If x be a quantity so small that x^3 may be neglected in comparison of a^3 , prove that

$$\sqrt{\frac{a}{a+x}} + \sqrt{\frac{a}{a-x}} = 2 + \frac{3x^2}{4a^2}.$$

109. If p be nearly equal to q , prove that

$$\left(\frac{5p+4q}{4p+5q}\right) = \left(\frac{p}{q}\right)^{1/9}.$$

110. If x be nearly equal to 1, prove that

$$\frac{mx^m - nx^n}{m-n} = x^{m+n}.$$

EXPONENTIAL SERIES

111. Find the co-efficient of x^n in the expansion of e^{5x} .

112. Find the co-efficient of x^n in the expansion of e^{5x+4} .

113. Find the co-efficient of x^n in $\left(\frac{1+3x+2x^2}{e^x}\right)$.

114. If $\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \dots + B_nx^n$, prove that

$$B_n - B_{n-1} = \frac{1}{n!}.$$

115. Find the value of

$$\frac{\left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\right)}{\left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots\right)}.$$

116. Find the value of $(x+1)^{x+1} + 10$, if

$$x = \left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\right)^2 - \left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots\right)^2.$$

117. If $a = \sum_{n=1}^{\infty} \left(\frac{2n}{(2n-1)!}\right)$, $b = \sum_{n=1}^{\infty} \left(\frac{2n}{(2n+1)!}\right)$, find the value of ab .

118. Find the sum of

$$1 + \frac{1+2}{1!} + \frac{1+2+3}{2!} + \frac{1+2+3+4}{3!} + \dots$$

119. Find the value of $\sum_{n=4}^{\infty} \left(\frac{C(n, 4)}{P(n, n)}\right) = \frac{e}{24}$.

120. If $a = \sum_{n=0}^{\infty} \left(\frac{x^{3n}}{(3n)!}\right)$, $b = \sum_{n=0}^{\infty} \left(\frac{x^{3n-2}}{(3n-2)!}\right)$,

$$\text{and } c = \sum_{n=0}^{\infty} \left(\frac{x^{3n-1}}{(3n-1)!}\right),$$

prove that $a^3 + b^3 + c^3 - 3abc = 1$.

LOGARITHMIC SERIES

121. Prove that $\log_e 2 < 1 < \log_e 3$.

122. Find the value of

$$\frac{1}{4} - \frac{1}{2} \left(\frac{1}{4}\right)^2 + \frac{1}{3} \left(\frac{1}{4}\right)^3 - \frac{1}{4} \left(\frac{1}{4}\right)^4 + \dots$$

123. Prove that

$$\log_e \left(\frac{x+1}{x}\right) = 2 \left(\frac{1}{(2x+1)} + \frac{1}{3(2x+1)^3} + \frac{1}{5(2x+1)^5} + \dots\right).$$

124. Find the co-efficient of x^n in the expansion of $\log_e(1 + 6x + 8x^2)$.

125. If $y = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$,

$$\text{prove that } x = y - \frac{y^2}{2!} + \frac{y^3}{3!} - \frac{y^4}{4!} + \dots$$

126. Prove that

$$\log_e(1 + x + x^2 + x^3 + x^4 + \dots) = \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \dots\right).$$

127. Find the sum of

$$\frac{1}{2.3} + \frac{1}{4.5} + \frac{1}{6.7} + \dots$$

128. Find the sum of

$$\frac{1}{1.2.3} + \frac{1}{3.4.5} + \frac{1}{5.6.7} + \dots$$

129. Find the sum of

$$1 + \left(\frac{1}{2} + \frac{1}{3}\right)\frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5}\right)\frac{1}{4^2} + \left(\frac{1}{6} + \frac{1}{7}\right)\frac{1}{4^3} + \dots$$

130. Find the sum of

$$1 + \left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right) + \left(\frac{3-2\sqrt{2}}{12}\right) + \left(\frac{5\sqrt{2}-7}{24\sqrt{2}}\right) + \left(\frac{17-12\sqrt{2}}{80}\right) + \dots$$

HINTS AND SOLUTIONS

LEVEL I

1. We have,

$$\begin{aligned}(x+1)^6 &= {}^6C_0x^6 + {}^6C_1x^5 + {}^6C_2x^4 + {}^6C_3x^3 \\ &\quad + {}^6C_4x^2 + {}^6C_5x + {}^6C_6x^0 \\ &= x^6 + 6x^5 + 15x^4 + 20x^3 + 15x^2 + 6x + 1\end{aligned}$$

2. We have,

$$\begin{aligned}\left(\frac{x}{5} + \frac{1}{x}\right)^5 &= {}^5C_0\left(\frac{x}{5}\right)^5\left(\frac{1}{x}\right)^0 + {}^5C_1\left(\frac{x}{5}\right)^4\left(\frac{1}{x}\right)^1 + {}^5C_2\left(\frac{x}{5}\right)^3\left(\frac{1}{x}\right)^2 \\ &\quad + {}^5C_3\left(\frac{x}{5}\right)^2\left(\frac{1}{x}\right)^3 + {}^5C_4\left(\frac{x}{5}\right)^1\left(\frac{1}{x}\right)^4 + {}^5C_5\left(\frac{x}{5}\right)^0\left(\frac{1}{x}\right)^5\end{aligned}$$

3. We have,

$$\begin{aligned}(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6 &= (x + a)^6 + (x - a)^6, \quad a = \sqrt{x^2 - 1} \\ &= 2({}^6C_0x^6a^0 + {}^6C_2x^4a^2 + {}^6C_4x^2a^4 + {}^6C_6x^0a^6) \\ &= 2({}^6C_0x^6 + {}^6C_2x^4a^2 + {}^6C_4x^2a^4 + {}^6C_6a^6) \\ &= 2[{}^6C_0x^6 + {}^6C_2x^4(x^2 - 1) \\ &\quad + {}^6C_4x^2(x^2 - 1)^2 + {}^6C_6(x^2 - 1)^3]\end{aligned}$$

Hence, the degree of the polynomial is 6.

4. We have, the number of terms in

(i) $(1+x)^{2013}$ is 2014.

(ii) $(1+2x+x^2)^{1007} = ((1+x)^2)^{1007} = (1+x)^{2014}$ is 2015.

(iii) $(1+3x+3x^2+x^3)^{668} = ((1+x)^3)^{668} = (1+x)^{2004}$ is 2005.

(iv) $(1+4x+6x^2+4x^3+x^4)^{503} = ((1+x)^4)^{503} = (1+x)^{2012}$ is 2013.

(v) $(1+x)(1+x^2)^{10}$ is 12.

(vi) $(1-x)^{11}(1+x+x^2)^{10}$
 $= (1-x)\{(1-x)(1+x+x^2)\}^{10}$
 $= (1-x)(1-x^3)^{10}$ is 12.

5. The total number of terms in the expansion of $(a+b)^{50} + (a-b)^{50} = \left(\frac{50}{2} + 1\right) = 26$.

6. The total number of terms in the expansion of $\left(x + \frac{1}{x}\right)^{99} + \left(x - \frac{1}{x}\right)^{99} = \left(\frac{99+1}{2}\right) = 50$.

7. The total number of terms in the expansion of $\left(x^2 + \frac{1}{x^2}\right)^{20} - \left(x^2 - \frac{1}{x^2}\right)^{20} = \left(\frac{20}{2}\right) = 10$

8. The total number of terms in the expansion of $\left(\frac{2}{x} + x^3\right)^{2015} - \left(\frac{2}{x} - x^3\right)^{2015} = \left(\frac{2015+1}{2}\right) = 1008$

9. We have

$$\begin{aligned}t_{13} &= t_{12+1} \\ &= {}^{18}C_{12} \times (9x)^{18-12} \times \left(-\frac{1}{3\sqrt{x}}\right)^{12} \\ &= {}^{18}C_{12} \times (9x)^6 \times \left(\frac{1}{3^{12}x^6}\right) \\ &= {}^{18}C_{12}\end{aligned}$$

10. We have, 4th term from the end = $(7+1) - (4-1)$
 $= 5$ th term from the beginning.

$$\begin{aligned}\text{Thus, } t_5 = t_{4+1} &= {}^7C_4 \times \left(\frac{3}{x^2}\right)^{7-4} \times \left(-\frac{x^3}{6}\right)^4 \\ &= {}^7C_4 \times \left(\frac{3}{x^2}\right)^3 \times \left(\frac{x^3}{6}\right)^4 \\ &= {}^7C_4 \times \left(\frac{x^6}{48}\right) = \frac{35}{48} \cdot x^6\end{aligned}$$

11. Let t_{r+1} th term be the independent of x .

$$\begin{aligned} t_{r+1} &= {}^6C_r \times \left(\frac{3x^2}{2}\right)^{6-r} \times \left(-\frac{1}{3x}\right)^r \\ &= {}^6C_r \times \left(\frac{3}{2}\right)^{6-r} \times \left(-\frac{1}{3}\right)^r \times x^{12-2r-r} \end{aligned}$$

Since the term is independent of x , so

$$12 - 3r = 0, \text{ i.e. } r = 4$$

Thus, the 5th term is the independent of x .

12. Let t_{r+1} term has the co-efficient of x^7 .

$$\begin{aligned} \text{Now, } t_{r+1} &= {}^{11}C_r \times (3x^2)^{11-r} \times \left(\frac{1}{5x}\right)^r \\ &= {}^{11}C_r \times (3)^{11-r} \times \left(\frac{1}{5}\right)^r \times x^{22-2r-r} \end{aligned}$$

$$\therefore 22 - 3r = 7$$

$$\Rightarrow 3r = 15$$

$$\Rightarrow r = 5$$

Thus, the co-efficient of x^7 is ${}^{11}C_5 \times 3^6 \times \left(\frac{1}{5}\right)^5$.

13. Given,

$$t_3 = 10000$$

$$\Rightarrow {}^5C_2 (x)^{5-2} (x^{\log_{10} x})^2 = 10000$$

$$\Rightarrow 10 \cdot (x)^3 (x^{2\log_{10} x}) = 10000$$

$$\Rightarrow (x)^{3+2\log_{10} x} = 1000$$

$$\Rightarrow (x)^{3+2\log_{10} x} = 10^3$$

$$\Rightarrow 3 + 2 \log_{10} x = \log_{10^3} x = \frac{1}{3} \log_{10} x$$

$$\Rightarrow \left(2 - \frac{1}{3}\right) \log_{10} x = -3$$

$$\Rightarrow \log_{10} x = -\frac{9}{5}$$

$$\Rightarrow x = 10^{-9/5}$$

14. We have, the co-efficient of x^n in

$$\begin{aligned} (1+x)^{2n} &= {}^{2n}C_n \\ &= \frac{(2n)!}{(n)! \times (n)!} \\ &= \frac{2n \times (2n-1)!}{(n)! \times (n)!} \\ &= \frac{2 \times (2n-1)!}{(n)! \times (n-1)!} \\ &= 2 \times \frac{(2n-1)!}{(n)! \times (n-1)!} \\ &= 2 \times {}^{2n-1}C_n \\ &= 2 \text{ times the co-efficient of } x^n \text{ in } (1+x)^{2n-1}. \end{aligned}$$

15. Since,

$${}^{2n}C_1, {}^{2n}C_2, {}^{2n}C_3 \in \text{AP}$$

$$\Rightarrow {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$$

$$\Rightarrow 2 \cdot \left(\frac{2n(2n-1)}{2}\right) = 2n + \left(\frac{2n(2n-1)(2n-2)}{6}\right)$$

$$\Rightarrow 2n(2n-1) = 2n + \left(\frac{2n(2n-1)(2n-2)}{6}\right)$$

$$\Rightarrow (2n-1) = 1 + \frac{(2n-1)(2n-2)}{6}$$

$$\Rightarrow (12n-6) = 6 + (2n-1)(2n-2)$$

$$\Rightarrow (12n-6) = 6 + (4n^2 - 6n + 2)$$

$$\Rightarrow (4n^2 - 18n + 14) = 0$$

$$\Rightarrow (2n^2 - 9n + 7) = 0$$

16. Since,

${}^nC_4, {}^nC_5, {}^nC_6$ are in AP

$$\Rightarrow {}^{2n}C_5 = {}^{2n}C_4 + {}^{2n}C_6$$

$$\Rightarrow 2 \frac{n!}{5! \times (n-5)!} = \frac{n!}{4! \times (n-4)!} + \frac{n!}{6! \times (n-6)!}$$

$$\Rightarrow \frac{2}{5 \times (n-5)} = \frac{1}{(n-4) \times (n-5)} + \frac{1}{6 \times 5}$$

$$\Rightarrow \frac{2}{5 \times (n-5)} - \frac{1}{(n-4) \times (n-5)} = \frac{1}{6 \times 5}$$

$$\Rightarrow \frac{2(n-4) - 5}{5 \times (n-4) \times (n-5)} = \frac{1}{6 \times 5}$$

$$\Rightarrow \frac{2n-13}{(n-4) \times (n-5)} = \frac{1}{6}$$

$$\Rightarrow 6(2n-13) = (n-4) \times (n-5)$$

$$\Rightarrow 12n - 78 = n^2 - 9n + 20$$

$$\Rightarrow n^2 - 21n + 98 = 0$$

$$\Rightarrow (n-7)(n-14) = 0$$

$$\Rightarrow n = 14, 7$$

Hence, the values of n are 7 and 14.

- 17 We have

$$t_2 = {}^nC_1 x, t_3 = {}^nC_2 x^2, t_4 = {}^nC_3 x^3$$

Since ${}^nC_1, {}^nC_2, {}^nC_3 \in \text{AP}$

$$\Rightarrow {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$$

$$\Rightarrow 2 \cdot \frac{n(n-1)}{2} = n + \frac{n(n-1)(n-2)}{6}$$

$$\Rightarrow n(n-1) = n + \frac{n(n-1)(n-2)}{6}$$

$$\Rightarrow 6n(n-1) = 6n + n(n-1)(n-2)$$

$$\Rightarrow 6n^2 - 6n = 6n + n(n^2 - 3n + 2)$$

$$\Rightarrow 6n^2 - 6n = 6n + n^3 - 3n^2 + 2n$$

$$\Rightarrow n^3 - 9n^2 + 14n = 0$$

$$\Rightarrow n(n^2 - 9n + 14) = 0$$

$$\Rightarrow n(n-2)(n-7) = 0$$

$$\Rightarrow n = 0, 2, 7$$

$$\Rightarrow n = 7$$

18. The co-efficient of x^{10} in $(1+x)^{50}$ is ${}^{50}C_{10}$.

19. We have,

$$(1+x^2)^{30}$$

$$= 1 + {}^{30}C_1 x^2 + {}^{30}C_2 (x^2)^2 + {}^{30}C_3 (x^2)^3 + \dots + {}^{30}C_5 (x^2)^5 + \dots + {}^{30}C_{30} (x^2)^{30}$$

Thus, the co-efficient of x^{10} is ${}^{30}C_5$.

20. We have,

$$\begin{aligned} & (1+x)^{41}(1-x+x^2)^{40} \\ &= (1+x) \times \{(1+x)(1-x+x^2)\}^{40} \\ &= (1+x) \times (1+x^3)^{40} \\ &= (1+x) \times [1 + {}^{40}C_1 x^3 + {}^{40}C_2 (x^3)^2 + \dots + {}^{40}C_{16} (x^3)^{16} + {}^{40}C_{17} (x^3)^{17} + \dots] \end{aligned}$$

Thus, the co-efficient of x^{50} is 0.

21. We have,

$$\begin{aligned} & (1+x)^5 \times (1-x)^6 \\ &= (1 + {}^5C_1 x + {}^5C_2 x^2 + {}^5C_3 x^3 + \dots) \\ & \quad \times (1 - {}^6C_1 x + {}^6C_2 x^2 - {}^6C_3 x^3 + \dots) \end{aligned}$$

Thus, co-efficient of $x = {}^5C_1 - {}^6C_1 = 5 - 6 = -1$

The co-efficient of $x^2 = {}^6C_2 + {}^5C_2 - {}^6C_1 \times {}^5C_1$
 $= 15 + 10 - 30 = -5$

The co-efficient of x^3

$$\begin{aligned} &= {}^5C_3 - {}^6C_3 + {}^5C_1 \times {}^6C_2 - {}^6C_1 \times {}^5C_2 \\ &= 10 - 20 + 5 \cdot 15 - 6 \cdot 10 \\ &= 10 - 20 + 75 - 60 \\ &= 5 \end{aligned}$$

22. We have,

$$\begin{aligned} & (1+x)^m (1-x)^n \\ &= (1 + {}^mC_1 x + {}^mC_2 x^2 + \dots) \times (1 - {}^nC_1 x + {}^nC_2 x^2 - \dots) \end{aligned}$$

The co-efficient of $x = 3$

$$\begin{aligned} \Rightarrow & {}^mC_1 - {}^nC_1 = 3 \\ \Rightarrow & m - n = 3 \end{aligned} \quad \dots(i)$$

The co-efficient of $x^2 = -6$

$$\begin{aligned} \Rightarrow & {}^mC_2 + {}^nC_2 - {}^mC_1 \times {}^nC_1 = -6 \\ \Rightarrow & \frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn = -6 \\ \Rightarrow & m(m-1) + n(n-1) - 2mn = -12 \\ \Rightarrow & m^2 + n^2 - 2mn - (m+n) = -12 \\ \Rightarrow & (m-n)^2 - (m+n) = -12 \\ \Rightarrow & (-3)^2 - (m+n) = -12 \\ \Rightarrow & -(m+n) = -21 \\ \Rightarrow & (m+n) = 21 \end{aligned} \quad \dots(ii)$$

Solving Eqs (i) and (ii), we get

$$m = 12 \text{ and } n = 9$$

Hence, the values of m and n are 12 and 9, respectively.

23. Since n is even, so the middle term $= \left(\frac{12}{2} + 1\right) = 7$ th term.

$$\begin{aligned} \text{Thus, } t_7 = t_{6+1} &= {}^{12}C_6 \times \left(\frac{2x^2}{3}\right)^{12-6} \times \left(-\frac{3}{2x}\right)^6 \\ &= {}^{12}C_6 \times x^6 \end{aligned}$$

24. Since n is odd, so, the middle terms are

$$\begin{aligned} & \left(\frac{9+1}{2}\right)\text{th and } \left(\frac{9+3}{2}\right)\text{th terms} \\ &= 5\text{th and } 6\text{th terms.} \end{aligned}$$

$$\begin{aligned} \text{Thus, } t_5 = t_{4+1} &= {}^9C_4 \times (3x)^{9-4} \times \left(-\frac{x^3}{6}\right)^4 \\ &= {}^9C_4 \times \left(\frac{3}{16}\right) \times x^{17} \\ \text{and } t_6 = t_{5+1} &= {}^9C_5 \times (3x)^{9-5} \times \left(-\frac{x^3}{6}\right)^5 \\ &= {}^9C_5 \times \left(-\frac{x^{19}}{96}\right) \end{aligned}$$

25. Since n is even, so the middle term is $(n+1)$ th term.

$$\begin{aligned} \text{Thus, } t_{n+1} &= {}^{2n}C_n \times x^n \\ &= \frac{(2n)!}{(n)! \times (n)!} \times x^n \\ &= \frac{2^n \times (n)! \times \{1.3.5 \dots (2n-1)\}}{(n)! \times (n)!} \times x^n \\ &= \frac{2^n \times \{1.3.5 \dots (2n-1)\}}{(n)!} \times x^n \end{aligned}$$

26. The sum of the co-efficients of two middle terms in $(1+x)^{2n-1}$

$$\begin{aligned} &= {}^{2n-1}C_{n-1} + {}^{2n-1}C_n \\ &= {}^{2n-1}C_n + {}^{2n-1}C_{n-1} \\ &= (2n-1+1)C_n \\ &= 2nC_n \\ &= \text{the co-efficient of the middle term in } (1+x)^{2n}. \end{aligned}$$

27. Since n is even, so the greatest co-efficient

$$\begin{aligned} &= {}^{100}C_{100/2} \times \left(\frac{1}{2}\right)^{100/2} \times \left(\frac{2}{3}\right)^{100/2} \\ &= {}^{100}C_{50} \times \left(\frac{1}{2}\right)^{50} \times \left(\frac{2}{3}\right)^{50} \\ &= {}^{100}C_{50} \times \left(\frac{1}{3}\right)^{50} \end{aligned}$$

28. Put $a = 1 = b$, $\Rightarrow 2^n = 4096 = 2^{12}$

$$\Rightarrow n = 12$$

Thus, the greatest co-efficient is ${}^{12}C_6$.

29. Put $x = 1$, $\Rightarrow 3^n = 6561 = 3^9$

$$\Rightarrow n = 9$$

Thus, the greatest co-efficients are

$$24 \times {}^9C_4 \text{ and } 2^5 \times {}^9C_5.$$

30. We have,

$$\begin{aligned} m &= \left| \frac{(n+1)|x|}{a+|x|} \right| \\ &= \left| \frac{(10+1)\left(\frac{3}{2} \times \frac{3}{5}\right)}{1 + \left(\frac{3}{2} \times \frac{3}{5}\right)} \right| = \frac{99}{19} = 5\frac{4}{19} \end{aligned}$$

$\Rightarrow m \neq$ not an integer

Thus, the greatest term = $t_{\left[\frac{5.4}{19}\right]+1}$
 $= t_6 = 6\text{th term.}$

31. We have,

$$m = \frac{(n+1)|x|}{a+|x|}$$

$$= \frac{(15+1)\left(-\frac{5}{3} \times \frac{1}{5}\right)}{1 + \left(-\frac{5}{3} \times \frac{1}{5}\right)} = \frac{\frac{16}{3}}{\frac{4}{3}} = 4$$

$\Rightarrow m =$ an integer

Thus, the greatest term are t_4 and t_{4+1} , i.e 4th and 5th terms.

32. We have,

$$(4^n - 3n - 1)$$

$$= (1 + 3)^n - 3n - 1$$

$$= (1 + {}^nC_1 \cdot 3 + {}^nC_2 \cdot 3^2 + \dots + {}^nC_n \cdot 3^n) - 3n - 1$$

$$= (1 + 3n + {}^nC_2 \cdot 3^2 + \dots + {}^nC_n \cdot 3^n) - 3n - 1$$

$$= ({}^nC_2 \cdot 3^2 + {}^nC_3 \cdot 3^3 + \dots + {}^nC_n \cdot 3^n)$$

$$= 3^2({}^nC_2 + {}^nC_3 \cdot 3 + \dots + {}^nC_n \cdot 3^{n-2})$$

Thus, $(4^n - 3n - 1)$ is divisible by 9.

33. We have,

$$3^{2n+2} - 8n - 9$$

$$= 3^2 \cdot 3^{2n} - 8n - 9$$

$$= 9 \cdot 9^n - 8n - 9$$

$$= 9(1 + 8)^n - 8n - 9$$

$$= 9(1 + {}^nC_1 \cdot 8 + {}^nC_2 \cdot 8^2$$

$$+ {}^nC_3 \cdot 8^3 + \dots + {}^nC_n \cdot 8^n) - 8n - 9$$

$$= 9(8n + {}^nC_2 \cdot 8^2 + {}^nC_3 \cdot 8^3$$

$$+ \dots + {}^nC_n \cdot 8^n) - 8n$$

$$= (8^2 n + {}^nC_2 \cdot 8^2 \cdot 9 + {}^nC_3 \cdot 8^3 \cdot 9$$

$$+ \dots + {}^nC_n \cdot 8^n \cdot 9)$$

$$= 8^2(n + {}^nC_2 \cdot 9 + {}^nC_3 \cdot 8 \cdot 9$$

$$+ \dots + {}^nC_n \cdot 8^{n-2} \cdot 9)$$

Thus, it is divisible by 64.

34. We have,

$$11^{n+2} + 12^{2n+1}$$

$$= 11^2 \cdot 11^n + 12 \cdot 12^{2n}$$

$$= 121 \cdot 11^n + 12 \cdot (144)^n$$

$$= 121 \cdot 11^n + 12 \cdot (11 + 133)^n$$

$$= 121 \cdot 11^n + 12[11^n + {}^nC_1 \cdot (11)^{n-1} 133$$

$$+ {}^nC_2 \cdot (11)^{n-2} 133^2 + \dots + {}^nC_n \cdot (11)^{n-n} 133^n]$$

$$= 133[11^n + {}^nC_1 \cdot (11)^{n-1} + \dots + {}^nC_n \cdot 133^{n-1}]$$

Thus, it is divisible by 133.

35. We have,

$$6^{n+2} + 7^{2n+1}$$

$$= 6^n \cdot 6^2 + 7^{2n} \cdot 7$$

$$= 36 \cdot 6^n + 7(6 + 43)^n$$

$$= 36 \cdot 6^n + 7[6^n + {}^nC_1 \cdot (6)^{n-1} 43 +$$

$${}^nC_2 \cdot (6)^{n-2} 43^2 + \dots + {}^nC_n \cdot (6)^{n-n} 43^n]$$

$$= 43[6^n + {}^nC_1 \cdot (6)^{n-1} + \dots + {}^nC_n \cdot 43^{n-1}]$$

Thus, it is divisible by 43.

36. We have,

$$11^{10} - 1$$

$$= (1 + 10)^{10} - 1$$

$$= (1 + {}^{10}C_1 \cdot 10 + {}^{10}C_2 \cdot 10^2 + {}^{10}C_3 \cdot 10^3$$

$$+ \dots + {}^{10}C_{100} \cdot 10^{100}) - 1$$

$$= ({}^{10}C_1 \cdot 10 + {}^{10}C_2 \cdot 10^2 + {}^{10}C_3 \cdot 10^3$$

$$+ \dots + {}^{10}C_{100} \cdot 10^{100})$$

$$= 100({}^{10}C_2 + {}^{10}C_3 \cdot 10 + \dots + {}^{10}C_{100} \cdot 10^{98})$$

Thus, it is divisible by 100.

37. We have,

$$7^{98}$$

$$= (7^2)^{49}$$

$$= (50 - 1)^{49}$$

$$= (50^{49} - {}^{49}C_1 \cdot 50^{48} + {}^{49}C_2 \cdot 50^{47}$$

$$- {}^{49}C_3 \cdot 50^{46} + \dots - {}^{49}C_{49} \cdot 1)$$

$$= (50^{49} - {}^{49}C_1 \cdot 50^{48} + {}^{49}C_2 \cdot 50^{47}$$

$$- {}^{49}C_3 \cdot 50^{46} + \dots - 5) + 4$$

Thus, the remainder is 4.

38. We have

$$1^{2013} + 2^{2013} + 3^{2013} + \dots + 2011^{2013} + 2012^{2013}$$

$$= (1^{2013} + 2012^{2013}) + (2^{2013} + 2011^{2013})$$

$$+ \dots + (1006^{2013} + 1007^{2013})$$

Here, each bracket is of the form of $(a^{2n+1} + b^{2n+1})$ and so is divisible by $(a + b)$.

Hence, the given expression is divisible by 2013.

39. We have

$$1992^{1998} - 1955^{1998} - 1938^{1998} + 1901^{1998}$$

$$= (1992^{1998} - 1938^{1998}) - (1955^{1998} - 1901^{1998})$$

Clearly, it is divisible by 54.

Also, the given expression can be written as

$$(1992^{1998} - 1955^{1998}) - (1938^{1998} - 1901^{1998})$$

It is divisible by 37.

Thus, the given expression

$$1992^{1998} - 1955^{1998} - 1938^{1998} + 1901^{1998}$$

is divisible by $54 \times 37 = 1998$

40. We have,

$$53^{53} - 33^3$$

$$= ((53^{53} - 43^{53}) + (43^{53} - 33^3))$$

$$= ((53^{53} - 43^{53}) + (43^{53} - 33^{43}) + (33^{43} - 33^3))$$

Clearly, each bracket is divisible by 10.

$$\begin{aligned}
 \text{Now, } (33^{43} - 33^3) &= 33^3(33^{40} - 1) \\
 &= 33^3((1 + 32)^{40} - 1) \\
 &= 33^3((1 + {}^{40}C_1 \cdot 32 + {}^{40}C_2 \cdot 32^2 \\
 &\quad + \dots + {}^{40}C_{40} \cdot 32^{40}) - 1) \\
 &= 33^3({}^{40}C_1 \cdot 32 + {}^{40}C_2 \cdot 32^2 \\
 &\quad + \dots + {}^{40}C_{40} \cdot 32^{40})
 \end{aligned}$$

Here, it is divisible by 10.

41. We have $(27)^{50} + (18)^{50}$.

The unit digit of $(27)^{50}$ is 9.

$$\begin{aligned}
 \text{The unit digit of } (18)^{50} &= (2 \times 3^2)^{50} \\
 &= (2^{50} \times 3^{100}) \\
 &= (2^{(4 \times 12 + 2)} \times 3^{4 \times 25}) \text{ is } 4 \times 1
 \end{aligned}$$

Thus, the unit digit of $(27)^{50} + (18)^{50}$ is unit digit of $(9 + 4)$ i.e. 3.

42. We know that $5! + 6! + 7! + \dots + (33)!$ is divisible by 10.

Now, the unit digit of

$1! + 2! + 3! + 4! + 5! + \dots + (33)!$ is the unit digit of $1! + 2! + 3! + 4!$.

Thus, the unit digit of $1! + 2! + 3! + 4!$ is the unit digit of $(1 + 2 + 6 + 24) (= 33)$ is 3.

43. We have,

$$\begin{aligned}
 (27)^{27} &= (3)^{81} = 3(3)^{80} = 3(9)^{40} = 3(81)^{20} \\
 &= 3(80 + 1)^{20} = 3(1 + 80)^{20} \\
 &= 3[1 + {}^{20}C_1 \cdot 80 + {}^{20}C_2 \cdot 80^2 + {}^{20}C_3 \cdot 80^3 + \\
 &\quad + \dots + {}^{20}C_{19} \cdot 80^{19} + {}^{20}C_{20} \cdot 80^{20}] \\
 &= 3(1 + 1600) + \text{a multiple of } 100 \\
 &= 4803 + \text{a multiple of } 100
 \end{aligned}$$

Clearly, last two digits = 03.

44. We have,

$$\begin{aligned}
 3^{999} &= 3(3^{998}) = 3(9^{499}) \\
 &= 3 \times 9(9^{498}) = 27(81^{249}) \\
 &= 27(1 + 80)^{249} \\
 &= 27[1 + {}^{249}C_1 \cdot 80 + {}^{249}C_2 \cdot 80^2 \\
 &\quad + {}^{249}C_3 \cdot 80^3 + \dots + {}^{249}C_{249} \cdot 80^{249}] \\
 &= 27(1 + {}^{249}C_1 \cdot 80) + 27[{}^{249}C_2 \cdot 80^2 \\
 &\quad + {}^{249}C_3 \cdot 80^3 + \dots + {}^{249}C_{249} \cdot 80^{249}] \\
 &= 27(1 + 249 \times 80) + \text{a multiple of } 100 \\
 &= 27 \times 19921 + \text{a multiple of } 100 \\
 &= 537867 + \text{a multiple of } 100
 \end{aligned}$$

Thus, the last two digit = 67.

45. We have,

$$\begin{aligned}
 (17)^{10} &= (17^2)^5 \\
 &= (289)^5 \\
 &= (290 - 1)^5
 \end{aligned}$$

$$\begin{aligned}
 &= (290^5 - {}^5C_1 \cdot 290^4 + {}^5C_2 \cdot 290^3 \\
 &\quad - {}^5C_3 \cdot 290^2 + {}^5C_4 \cdot 290 - 1) \\
 &= (290^5 - {}^5C_1 \cdot 290^4 + {}^5C_2 \cdot 290^3 - {}^5C_3 \cdot 290^2 \\
 &\quad + (5 \times 290 - 1)) \\
 &= (5 \times 290 - 1) + \text{a multiple of } 100 \\
 &= 1449 + \text{a multiple of } 100
 \end{aligned}$$

Thus, the last two digit is 49.

46. We have,

$$\begin{aligned}
 t_{r+1} &= {}^{500}C_r \times (9^{1/4})^{500-r} \times (8^{1/6})^r \\
 &= {}^{500}C_r \times (9)^{\frac{500-r}{4}} \times (8)^{\frac{r}{6}} \\
 &= {}^{500}C_r \times (3)^{\frac{500-r}{2}} \times (2)^{\frac{r}{3}}
 \end{aligned}$$

Thus, $r = 0, 2, 4, 6, 8, \dots, 500$

Hence, the number of integral terms = $250 + 1 = 251$.

47. We have,

$$\begin{aligned}
 t_{r+1} &= {}^{1000}C_r \times (3^{1/5})^{1000-r} \times (2^{1/3})^r \\
 &= {}^{1000}C_r \times (3)^{\frac{1000-r}{5}} \times (2)^{\frac{r}{3}}
 \end{aligned}$$

where $r = 0, 1, 2, 3, \dots, 1000$.

The total number of terms = 1001

The given term will be rational if the indices of 3 and 5 are integers. Thus, the given term will be rational, when the values of r are 0, 15, 30, 45, ..., 990.

The total number of rational terms = $66 + 1 = 67$

Hence, the total number of irrational terms = $(1001 - 67) = 934$

48. We have,

$$\begin{aligned}
 t_{r+1} &= {}^{100}C_r \times (5^{1/8})^{100-r} \times (2^{1/6})^r \\
 &= {}^{100}C_r \times (5)^{\frac{100-r}{8}} \times (2)^{\frac{r}{6}}
 \end{aligned}$$

The above expression is rational, if

$$\left(\frac{100-r}{8}\right) \text{ and } \left(\frac{r}{6}\right) \text{ are integers.}$$

Clearly, the values of r are 12, 36, 60, 84.

Thus, the number of non-integral terms = $101 - 4 = 97$

49. We have,

$$\begin{aligned}
 t_{r+1} &= {}^{10}C_r (2^{1/2})^{10-r} (3^{1/5})^r \\
 &= {}^{10}C_r (2)^{\frac{10-r}{2}} (3)^{\frac{r}{5}}
 \end{aligned}$$

where $r = 0, 10$

When $r = 0$, $t_1 = {}^{10}C_0 (2)^5 (3)^0 = 32$

When $r = 10$, $t_{11} = {}^{10}C_{10} (2)^0 (3)^2 = 9$

Therefore the sum of the rational terms,

$$t_1 + t_{11} = 32 + 9 = 41$$

50. Given $\sqrt{Q} - P < 1$

$$\Rightarrow 0 < (\sqrt{Q} - P)^n < 1$$

Let $(\sqrt{Q} - P)^n = f'$, where $0 < f' < 1$

Now, $(\sqrt{P} + Q)^n - (\sqrt{P} - Q)^n = 1 + f - f'$

Since n is odd, the LHS of the above expression contains an even powers of $\sqrt{P}$.

Hence, the terms in LHS and I are integers.

Therefore, $f - f'$ is an integer.

But $0 < f < 1$ and $-1 < f' < 0$

$\Rightarrow 0 < f < 1$ and $-1 < -f' < 0$

$\Rightarrow -1 < f - f' < 1$

$\Rightarrow f - f' = 0$

$\Rightarrow f = f'$

Thus, $(I + f)f' = (\sqrt{P} + Q)^n (\sqrt{P} - Q)^n$

$$= \{(\sqrt{P} + Q)(\sqrt{P} - Q)\}^n$$

$$= (P - Q^2)^n$$

$$= k^n$$

51. Given $\sqrt{Q} - P < 1$

$\Rightarrow 0 < (\sqrt{Q} - P)^n < 1$

Let $(\sqrt{Q} - P)^n = f'$, where $0 < f' < 1$.

Now, $(\sqrt{P} + Q)^n + (\sqrt{P} - Q)^n = I + f + f'$
= Even integer

But $0 < f < 1$ and $0 < f' < 1$

$\Rightarrow 0 < f + f' < 2$

$\Rightarrow f + f' = 1$

$\Rightarrow f' = 1 - f$

Hence,

$$(I + f)(I - f) = (I + f)f'$$

$$= (\sqrt{P} + Q)^n \times (\sqrt{P} - Q)^n$$

$$= (P - Q^2)^n$$

$$= k^n$$

52 Let I and f denote the integral and fractional part of R , respectively.

Given $f = R - [R]$

Let $R = I + f = (5\sqrt{5} + 11)^{2n+1}$, $0 < f < 1$

Also, $f' = (5\sqrt{5} - 11)^{2n+1}$, $0 < f' < 1$

Now,

$$R - f' = (5\sqrt{5} + 11)^{2n+1} - (5\sqrt{5} - 11)^{2n+1}$$

= an even integer.

But $0 < f < 1$ and $0 < f' < 1$

$\Rightarrow 0 < f < 1$ and $-1 < -f' < 0$

$\Rightarrow -1 < f - f' < 1$

$\Rightarrow f - f' = 0$

$\Rightarrow f = f'$

Therefore, $Rf = Rf'$

$$= (5\sqrt{5} + 11)^{2n+1} \cdot (5\sqrt{5} - 11)^{2n+1}$$

$$= (125 - 121)^{2n+1}$$

$$= 4^{2n+1}$$

53. Let $x = I + f = (8 + 3\sqrt{7})^n$, where $0 < f < 1$,

and $f' = (8 - 3\sqrt{7})^n$, $0 < f' < 1$.

Now,

$$1 + f + f' = (8 + 3\sqrt{7})^n + (8 - 3\sqrt{7})^n$$

= an even integer.

But $0 < f < 1$ and $0 < f' < 1$

$\Rightarrow 0 < f + f' < 2$

$\Rightarrow f + f' = 1$

Therefore, $I + f + f' = 2k$

$\Rightarrow I + 1 = 2k$

$\Rightarrow I = 2k - 1$

$\Rightarrow [x] = 2k - 1$

Now,

$$x - x^2 + x[x] = x - x(x - [x])$$

$$= x - xf$$

$$= x(1 - f)$$

$$= xf'$$

$$= (8 + 3\sqrt{7})^n \cdot (8 - 3\sqrt{7})^n$$

$$= (64 - 63)^n$$

$$= 1^n = 1$$

54. Given $\alpha + \beta = (8 + 3\sqrt{7})^n$, where $0 < \beta < 1$.

Let $\beta' = (8 - 3\sqrt{7})^n$, where $0 < \beta' < 1$.

Now,

$$\alpha + \beta + \beta' = (8 + 3\sqrt{7})^n + (8 - 3\sqrt{7})^n$$

= an even integer.

Also, $0 < \beta < 1$ and $0 < \beta' < 1$

$\Rightarrow 0 < \beta + \beta' < 2$

$\Rightarrow \beta + \beta' = 1$

$\Rightarrow \beta' = 1 - \beta$

Thus,

$$(1 - \beta)(\alpha + \beta) = (8 - 3\sqrt{7})^n \cdot (8 + 3\sqrt{7})^n$$

$$= (64 - 63)^n$$

$$= 1^n = 1$$

55. Let $I + f = (3 + \sqrt{5})^5$, where $0 < f < 1$,

and $f' = (3 - \sqrt{5})^5$, where $0 < f' < 1$.

Now,

$$I + f + f' = (3 + \sqrt{5})^5 + (3 - \sqrt{5})^5$$

$$= 2({}^5C_0 \cdot 3^5 + {}^5C_2 \cdot 3^3 \cdot 5 + {}^5C_4 \cdot 3 \cdot 5^2)$$

$$= 2(243 + 10 \cdot 135 + 5 \cdot 75)$$

$$= 2(243 + 1350 + 275)$$

$$= 3936$$

Thus, the number just greater than the number $(3 + \sqrt{5})^5$ is $I + f + f' = 3936$.

56. The co-efficient of a^n in $(1 + a)^{m+n}$

$$= {}^{m+n}C_n$$

$$= {}^{m+n}C_{m+n-n}$$

$$= {}^{m+n}C_m$$

= the co-efficient of a^m in $(1 + a)^{m+n}$.

57. We have,

$$\begin{aligned} {}^{10}C_r &= {}^{10}C_{r+4} \\ \Rightarrow r + r + 4 &= 10 \\ \Rightarrow 2r &= 10 - 4 = 6 \\ \Rightarrow r &= 3 \end{aligned}$$

58. We have,

$$\begin{aligned} \frac{C_1}{C_0} + 2 \cdot \frac{C_2}{C_1} + 3 \cdot \frac{C_3}{C_2} + \dots + n \cdot \frac{C_n}{C_{n-1}} \\ = \frac{n}{1} + 2 \cdot \left(\frac{n-1}{2}\right) + 3 \cdot \left(\frac{n-2}{3}\right) + \dots + n \cdot \left(\frac{1}{n}\right) \\ = n + (n-1) + (n-2) + \dots + 3 + 2 + 1 \\ = 1 + 2 + 3 + \dots + n \\ = \frac{n(n+1)}{2} \end{aligned}$$

59. We have

$$\begin{aligned} \frac{{}^{15}C_1}{{}^{15}C_0} + 2 \cdot \frac{{}^{15}C_2}{{}^{15}C_1} + 3 \cdot \frac{{}^{15}C_3}{{}^{15}C_2} + \dots + 15 \cdot \frac{{}^{15}C_{15}}{{}^{15}C_{14}} \\ = \frac{15 \times (15+1)}{2} \\ = \frac{15 \times 16}{2} \\ = 120 \end{aligned}$$

60. We have,

$$\begin{aligned} \left(1 + \frac{C_1}{C_0}\right) \left(1 + \frac{C_2}{C_1}\right) \left(1 + \frac{C_3}{C_2}\right) \dots \left(1 + \frac{C_n}{C_{n-1}}\right) \\ = \left(1 + \frac{n}{1}\right) \left(1 + \frac{n-1}{2}\right) \left(1 + \frac{n-2}{3}\right) \dots \left(1 + \frac{1}{n}\right) \\ = \left(\frac{1+n}{1}\right) \left(\frac{1+n}{2}\right) \left(\frac{1+n}{3}\right) \dots \left(\frac{1+n}{n}\right) \\ = \frac{(n+1)^n}{\{1.2.3 \dots n\}} \\ = \frac{(n+1)^n}{(n)!} \end{aligned}$$

61. We have,

$$\begin{aligned} ({}^nC_0 + {}^nC_1)({}^nC_1 + {}^nC_2)({}^nC_2 + {}^nC_3) \dots ({}^nC_n + {}^nC_{n-1}) \\ \left(1 + \frac{{}^nC_1}{{}^nC_0}\right) \left(1 + \frac{{}^nC_2}{{}^nC_1}\right) \left(1 + \frac{{}^nC_3}{{}^nC_2}\right) \dots \left(1 + \frac{{}^nC_n}{{}^nC_{n-1}}\right) \\ \times C_0 C_1 C_2 \dots C_{n-1} \\ = \left(\frac{1+n}{1}\right) \left(\frac{1+n}{2}\right) \left(\frac{1+n}{3}\right) \dots \left(\frac{1+n}{n}\right) \times k \\ = \frac{(n+1)^n}{(n)!} \times k \end{aligned}$$

62. We have,

$${}^nC_r + {}^nC_{r-1}$$

$$\begin{aligned} &= \frac{n!}{r! \times (n-r)!} + \frac{n!}{(r-1)! \times (n-r+1)!} \\ &= \frac{n!}{(r-1)! \times (n-r)!} \left(\frac{1}{r} + \frac{1}{n-r+1}\right) \\ &= \frac{n!}{(r-1)! \times (n-r)!} \left(\frac{n-r+1+r}{r(n-r+1)}\right) \\ &= \frac{(n+1) \times n!}{r \times (r-1)! \times (n-r+1) \times (n-r)!} \\ &= \frac{(n+1)!}{r! \times (n-r+1)!} \\ &= {}^{n+1}C_r \end{aligned}$$

63. We have,

$$\begin{aligned} \binom{n}{r} + 2 \binom{n}{r-1} + \binom{n}{r-2} \\ = {}^nC_r + 2 {}^nC_{r-1} + {}^nC_{r-2} \\ = ({}^nC_r + {}^nC_{r-1}) + ({}^nC_{r-1} + {}^nC_{r-2}) \\ = ({}^{n+1}C_r + {}^{n+1}C_{r-1}) \\ = {}^{n+2}C_r \end{aligned}$$

64. We have,

$$\begin{aligned} \binom{n}{r} + 3 \binom{n}{r-1} + 3 \binom{n}{r-2} + \binom{n}{r-3} \\ = \left(\binom{n}{r} + \binom{n}{r-1}\right) + 2 \left(\binom{n}{r-1} + \binom{n}{r-2}\right) \\ + \left(\binom{n}{r-2} + \binom{n}{r-3}\right) \\ = \binom{n+1}{r} + 2 \binom{n+1}{r-1} + \binom{n+1}{r-2} \\ = \left(\binom{n+1}{r} + \binom{n+1}{r-1}\right) + \left(\binom{n+1}{r-1} + \binom{n+1}{r-2}\right) \\ = \binom{n+2}{r} + \binom{n+2}{r-1} \\ = \binom{n+3}{r} \end{aligned}$$

65. We have,

$$\begin{aligned} \binom{n}{r} + 4 \binom{n}{r-1} + 6 \binom{n}{r-2} + 4 \binom{n}{r-3} + \binom{n}{r-4} \\ = \left(\binom{n}{r} + \binom{n}{r-1}\right) + 3 \left(\binom{n}{r-1} + \binom{n}{r-2}\right) \\ + 3 \left(\binom{n}{r-2} + \binom{n}{r-3}\right) + \left(\binom{n}{r-3} + \binom{n}{r-4}\right) \\ = \binom{n+1}{r} + 3 \binom{n+1}{r-1} + 3 \binom{n+1}{r-2} + \binom{n+1}{r-3} \end{aligned}$$

$$= \binom{n+4}{r}$$

66. We have,

$$\begin{aligned} & {}^{47}C_4 + \sum_{j=1}^5 {}^{52-j}C_3 \\ &= {}^{47}C_4 + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3) \\ &= ({}^{47}C_4 + {}^{47}C_3) + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3) \\ &= ({}^{48}C_4 + {}^{48}C_3) + ({}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3) \\ &= ({}^{49}C_4 + {}^{49}C_3) + ({}^{51}C_3 + {}^{50}C_3) \\ &= ({}^{50}C_4 + {}^{50}C_3) + ({}^{51}C_3) \\ &= ({}^{51}C_4 + {}^{51}C_3) \\ &= {}^{52}C_4 \end{aligned}$$

67. We have,

$$\begin{aligned} & {}^mC_m + {}^{m+1}C_m + {}^{m+2}C_m + \dots + {}^nC_m \\ &= {}^{m+1}C_{m+1} + {}^{m+1}C_m + {}^{m+2}C_m + \dots + {}^nC_m \\ &= ({}^{m+1}C_{m+1} + {}^{m+1}C_m) + {}^{m+2}C_m + \dots + {}^nC_m \\ &= ({}^{m+2}C_{m+1} + {}^{m+2}C_m) + {}^{m+3}C_m + \dots + {}^nC_m \\ &= ({}^{m+3}C_{m+1} + {}^{m+3}C_m) + {}^{m+4}C_m + \dots + {}^nC_m \\ &= {}^{m+4}C_{m+1} + \dots + {}^nC_m \\ &= {}^{n+1}C_{m+1} \end{aligned}$$

68. ${}^{20}C_{20} + {}^{21}C_{20} + {}^{22}C_{20} + {}^{23}C_{20} + \dots + {}^{2014}C_{20}$
 $= {}^{2015}C_{21}$

69. ${}^{48}C_{47} + {}^{49}C_{47} + {}^{50}C_{47} + {}^{51}C_{47} + \dots + {}^{1005}C_{47}$
 $= ({}^{47}C_{47} + {}^{48}C_{47} + {}^{49}C_{47} + \dots + {}^{1005}C_{47}) - {}^{47}C_{47}$
 $= {}^{1006}C_{48} - 1$

70. We have,

$$\begin{aligned} & {}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + {}^{n-3}C_r + \dots + {}^rC_r \\ &= \text{Co-efficient of } x^r \text{ in the expansion of} \\ & (1+x)^n + (1+x)^{n-1} + (1+x)^{n-2} + \dots + (1+x)^r \\ &= (1+x)^r + (1+x)^{r+1} + (1+x)^{r+2} + \dots + (1+x)^n \\ &= (1+x)^r (1 + (1+x) + (1+x)^2 + \dots + (1+x)^{n-r}) \\ &= (1+x)^r \left(\frac{(1+x)^{n-r+1} - 1}{(1+x) - 1} \right) \\ &= \left(\frac{(1+x)^{n+1} - (1+x)^r}{x} \right) \end{aligned}$$

$$\begin{aligned} \therefore \text{Co-efficient of } x^r \text{ in } \left(\frac{(1+x)^{n+1} - (1+x)^r}{x} \right) \\ &= \text{Co-efficient of } x^{r+1} \text{ in } (1+x)^{n+1} - (1+x)^r \\ &= {}^{n+1}C_{r+1} \end{aligned}$$

71. ${}^{49}C_{10} + {}^{48}C_{10} + {}^{47}C_{10} + \dots + {}^{10}C_{10} = {}^{50}C_{11}$

72. ${}^{2014}C_{100} + {}^{2013}C_{100} + {}^{2012}C_{100} + \dots + {}^{100}C_{100} = {}^{2015}C_{101}$

73. ${}^{70}C_{20} + {}^{69}C_{20} + {}^{68}C_{20} + {}^{67}C_{20} + \dots + {}^{21}C_{20}$
 $= {}^{70}C_{20} + {}^{69}C_{20} + {}^{68}C_{20} + \dots + {}^{21}C_{20} + {}^{20}C_{20} - {}^{20}C_{20}$
 $= {}^{71}C_{21} - 1$

74. Let $S = {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + {}^{2n}C_3 + \dots + {}^{2n}C_n \dots (i)$

$$S = {}^{2n}C_{2n} + {}^{2n}C_{2n-1} + {}^{2n}C_{2n-2} + {}^{2n}C_{2n-3} + \dots + {}^{2n}C_{2n-n} \dots (ii)$$

and $S = {}^{2n}C_{2n} + {}^{2n}C_{2n-1} + {}^{2n}C_{2n-2} + {}^{2n}C_{2n-3} + \dots + C_n$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2S &= ({}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots \\ &\quad + {}^{2n}C_n + {}^{2n}C_{n+1} + \dots + {}^{2n}C_{2n}) = {}^{2n}C_n \\ 2S &= 2^{2n} + {}^{2n}C_n \\ S &= 2^{2n-1} + \frac{1}{2} \times {}^{2n}C_n \end{aligned}$$

Hence, the result.

75. The required sum $= 2^{29} + \frac{1}{2} \times {}^{30}C_{15}$

76. Required sum $= 2^{39} + \frac{1}{2} \times {}^{40}C_{20}$

77. We have,

$$\begin{aligned} & {}^nC_3 + 2 \cdot {}^{n+1}C_3 + 3 \cdot {}^{n+2}C_3 + \dots + n \cdot {}^{2n-1}C_3 \\ &= \text{Co-efficient of } x^3 \text{ in the expansion of} \\ & (1+x)^n + 2(1+x)^{n+1} + 3(1+x)^{n+2} \\ & \quad + \dots + n \cdot (1+x)^{2n-1} \end{aligned}$$

Let $S = (1+x)^n + 2(1+x)^{n+1} + 3(1+x)^{n+2} + \dots + n \cdot (1+x)^{2n-1} \dots (i)$

$\therefore (1+x)S = (1+x)^{n+1} + 2(1+x)^{n+2} + 3(1+x)^{n+3} + \dots + n \cdot (1+x)^{2n} \dots (ii)$

Subtracting Eqs (ii) from (i), we get

$$\begin{aligned} -xS &= (1+x)^n + (1+x)^{n+1} \\ &\quad + \dots + (1+x)^{2n-1} + (1+x)^{2n} - n(1+x)^{2n} \\ &= (1+x)^n (1 + (1+x) + \dots + (1+x)^{n-1}) - n(1+x)^{2n} \\ &= (1+x)^n \left(\frac{(1+x)^n - 1}{(1+x) - 1} \right) - n(1+x)^{2n} \\ &= \left(\frac{(1+x)^{2n} - (1+x)^n}{x} \right) - n(1+x)^{2n} \\ \Rightarrow S &= \left(\frac{(1+x)^n - (1+x)^{2n}}{x^2} \right) + \frac{n(1+x)^{2n}}{x} \end{aligned}$$

Hence, the co-efficient of x^3 in

$$\begin{aligned} & \left(\frac{(1+x)^n - (1+x)^{2n}}{x^2} \right) - \frac{n(1+x)^{2n}}{x} \\ &= \text{the co-efficient of } x^5 \text{ in} \\ & [(1+x)^n - (1+x)^{2n} - nx(1+x)^{2n}] \\ &= -{}^{2n}C_5 + {}^nC_5 + n \cdot {}^{2n}C_4 \end{aligned}$$

78. We have,

$$^{100}C_{50} + 2 \cdot ^{99}C_{49} + 3 \cdot ^{98}C_{48} + \dots + 51 \cdot ^{50}C_0$$

$$= ^{100}C_{50} + 2 \cdot ^{99}C_{50} + 3 \cdot ^{98}C_{50} + \dots + 51 \cdot ^{50}C_{50}$$

= Co-efficient of x^{50} in the expansion of

$$(1+x)^{100} + 2 \cdot (1+x)^{99} + 3 \cdot (1+x)^{98} + \dots + 51 \cdot (1+x)^{50}$$

Let $S = (1+x)^{100} + 2 \cdot (1+x)^{99} + 3 \cdot (1+x)^{98} + \dots + 51 \cdot (1+x)^{50}$... (i)

$\therefore \frac{S}{(1+x)} = (1+x)^{99} + 2 \cdot (1+x)^{98} + 3 \cdot (1+x)^{97} + \dots + 51 \cdot (1+x)^{49}$... (ii)

Subtracting Eqs (ii) from (i), we get

$$\left(1 - \frac{1}{1+x}\right)S = (1+x)^{100} + (1+x)^{99} + \dots + (1+x)^{50} - 51(1+x)^{49}$$

$$\left(\frac{x}{1+x}\right)S = (1+x)^{100} + (1+x)^{99} + \dots + (1+x)^{50} - 51(1+x)^{49}$$

$$= (1+x)^{50} \left(\frac{(1+x)^{51} - 1}{(1+x) - 1} \right) - 51(1+x)^{49}$$

$$= \left(\frac{(1+x)^{101} - (1+x)^{50}}{x} \right) - 51(1+x)^{49}$$

$$\Rightarrow S = \left(\frac{(1+x)^{102} - (1+x)^{51}}{x^2} \right) - 51 \frac{(1+x)^{50}}{x}$$

Hence, the co-efficient of x^{50} in

$$\left(\frac{(1+x)^{102} - (1+x)^{51}}{x^2} \right) - 51 \frac{(1+x)^{50}}{x}$$

= the co-efficient of x^{52} in

$$[(1+x)^{102} - (1+x)^{51} - 51x(1+x)^{50}]$$

$$= ^{102}C_{52}$$

79. We have

$$(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000}$$

$$= (1+x)^{1000} \left(1 + 2 \cdot \left(\frac{x}{1+x} \right) + 3 \left(\frac{x}{1+x} \right)^2 + \dots + 1001 \cdot \left(\frac{x}{1+x} \right)^{1000} \right)$$

... (i)

Let

$$S = \left(1 + 2 \cdot \left(\frac{x}{1+x} \right) + 3 \left(\frac{x}{1+x} \right)^2 + \dots + 1001 \cdot \left(\frac{x}{1+x} \right)^{1000} \right)$$

$$r = \left(\frac{x}{1+x} \right)$$

$$S = (1 + 2 \cdot r + 3 \cdot r^2 + \dots + 1001 \cdot r^{1000}), \quad \dots (ii)$$

$$\therefore r \cdot S = (r + 2 \cdot r^2 + 3 \cdot r^3 + \dots + 1001 \cdot r^{1001}) \quad \dots (iii)$$

Subtracting Eqs (iii) from (ii), we get

$$(1-r) \cdot S = (1 + r + r^2 + r^3 + \dots + r^{1000}) - 1001r^{1001}$$

$$= \left(\frac{1-r^{1001}}{1-r} \right) - 1001r^{1001}$$

$$\Rightarrow S = \left(\frac{1-r^{1001}}{(1-r)^2} \right) - 1001 \frac{r^{1001}}{(1-r)}$$

$$= \left(\frac{1 - \left(\frac{x}{1+x} \right)^{1001}}{\left(1 - \frac{x}{1+x} \right)^2} \right) - 1001 \frac{\left(\frac{x}{1+x} \right)^{1001}}{\left(1 - \frac{x}{1+x} \right)}$$

$$= (1+x)^{1000} \left(\frac{1 - \left(\frac{x}{1+x} \right)^{1001}}{\left(1 - \frac{x}{1+x} \right)^2} \right) - 1001 \frac{\left(\frac{x}{1+x} \right)^{1001}}{\left(1 - \frac{x}{1+x} \right)}$$

$$= \left(\frac{(1+x)^{1000} - \left(\frac{x^{1001}}{1+x} \right)}{\left(\frac{1}{1+x} \right)^2} \right) - 1001 \frac{\left(\frac{x^{1001}}{1+x} \right)}{\left(\frac{1}{1+x} \right)}$$

$$= (1+x)^{1002} - x^{1001}(1+x) - 1001x^{1001}$$

Thus, the co-efficient of x^{50} in the expn of

$$[(1+x)^{1002} - x^{1001}(1+x) - 1001x^{1001}]$$

$$= ^{1002}C_{50}$$

80. Let $t_r = r \cdot C_r$

$$= r \cdot {}^nC_r$$

$$= r \times \frac{n}{r} \times {}^{n-1}C_{r-1} = n \cdot {}^{n-1}C_{r-1}$$

Thus, $S_n = \sum_{r=1}^n t_r = \sum_{r=1}^n (n \cdot {}^{n-1}C_{r-1})$

$$= n \sum_{r=1}^n ({}^{n-1}C_{r-1}) = n 2^{n-1}$$

81. Let $t_{r+1} = (r+1) \cdot C_r = (r+1) \cdot {}^nC_r$

$$= r \cdot {}^nC_r + {}^nC_r$$

$$= r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} + {}^nC_r$$

$$= n \cdot {}^{n-1}C_{r-1} + {}^nC_r$$

$$\begin{aligned}
 \text{Thus, } S_n &= \sum_{r=0}^n t_{r+1} = \sum_{r=0}^n (n \cdot {}^{n-1}C_{r-1} + {}^nC_r) \\
 &= n \sum_{r=0}^n {}^{n-1}C_{r-1} + \sum_{r=0}^n {}^nC_r \\
 &= n \cdot 2^{n-1} + 2^n \\
 &= (n+2)2^{n-1}
 \end{aligned}$$

$$\begin{aligned}
 82. \text{ Let } t_r &= r^2 \cdot C_r \\
 &= r^2 \cdot {}^nC_r \\
 &= \{r(r-1) + r\} \cdot {}^nC_r \\
 &= \{r(r-1) {}^nC_r + r {}^nC_r\} \\
 &= \left\{ r(r-1) \times \left(\frac{n(n-1)}{r(r-1)} \right) {}^{n-2}C_{r-2} \right. \\
 &\quad \left. + r \left(\frac{n}{r} \right) {}^{n-1}C_{r-1} \right\} \\
 &= \{n(n-1) {}^{n-2}C_{r-2} + n {}^{n-1}C_{r-1}\}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } S_n &= \sum_{r=1}^n t_r \\
 &= \sum_{r=1}^n (n(n-1) {}^{n-2}C_{r-2} + n {}^{n-1}C_{r-1}) \\
 &= n(n-1) \sum_{r=1}^n {}^{n-2}C_{r-2} + \sum_{r=1}^n n {}^{n-1}C_{r-1} \\
 &= n(n-1) \cdot 2^{n-2} + n \cdot 2^{n-1} \\
 &= n[(n-1) \cdot 2^{n-2} + 2^{n-1}] \\
 &= n \cdot 2^{n-2}(n-1+2) \\
 &= n(n-1)2^{n-2}
 \end{aligned}$$

$$\begin{aligned}
 83. \text{ Let } t_{r+1} &= \frac{C_r}{r+1} = \frac{{}^nC_r}{r+1} \\
 &= \frac{1}{n+1} \left(\frac{n+1}{r+1} \times {}^nC_r \right) \\
 &= \frac{{}^{n+1}C_{r+1}}{n+1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } S_n &= \sum_{r=0}^n t_{r+1} = \sum_{r=0}^n \left(\frac{{}^{n+1}C_{r+1}}{n+1} \right) \\
 &= \frac{1}{n+1} \times \sum_{r=0}^n ({}^{n+1}C_{r+1}) \\
 &= \frac{1}{n+1} \times (2^{n+1} - {}^{n+1}C_0) \\
 &= \frac{2^{n+1} - 1}{n+1}
 \end{aligned}$$

$$84. \text{ Let } t_{r+1} = \frac{C_r}{(r+1)(r+2)} = \frac{{}^nC_r}{(r+1)(r+2)}$$

$$\begin{aligned}
 &= \frac{1}{(n+1)(n+2)} \times \frac{(n+1)(n+2)}{(r+1)(r+2)} \times {}^nC_r \\
 &= \frac{{}^{n+2}C_{r+2}}{(n+1)(n+2)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } S_n &= \sum_{r=0}^n t_{r+1} = \sum_{r=0}^n \frac{{}^{n+2}C_{r+2}}{(n+1)(n+2)} \\
 &= \frac{1}{(n+1)(n+2)} \sum_{r=0}^n ({}^{n+2}C_{r+2}) \\
 &= \frac{2^{n+2} - {}^{n+2}C_0 - {}^{n+2}C_1}{(n+1)(n+2)} \\
 &= \frac{2^{n+2} - 1 - (n+2)}{(n+1)(n+2)} \\
 &= \frac{2^{n+2} - n - 3}{(n+1)(n+2)}
 \end{aligned}$$

85. We have,

$$\begin{aligned}
 (1+x)^m &= {}^mC_0 + {}^mC_1 \cdot x + {}^mC_2 \cdot x^2 + \\
 &\quad \dots + {}^mC_r \cdot x^r + {}^mC_r \cdot x^r + \dots + {}^mC_m \cdot x^m \quad \dots(i)
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } (1+x)^n &= {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \\
 &\quad \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \quad \dots(ii)
 \end{aligned}$$

Multiplying Eqs (i) and (ii), we get,

$$\begin{aligned}
 (1+x)^{m+n} &= ({}^mC_r \cdot {}^nC_0 + {}^mC_{r-1} \cdot {}^nC_1 \\
 &\quad + {}^mC_{r-2} \cdot {}^nC_2 + \dots + {}^mC_0 \cdot {}^nC_r) x^r \\
 &\quad + (\dots)x + (\dots)x^2 + \dots
 \end{aligned}$$

Comparing the co-efficients of x^r from both the sides, we get

$$\begin{aligned}
 &{}^mC_r \cdot {}^nC_0 + {}^mC_{r-1} \cdot {}^nC_1 + {}^mC_{r-2} \cdot {}^nC_2 \\
 &\quad + \dots + {}^mC_0 \cdot {}^nC_r = {}^{m+n}C_r
 \end{aligned}$$

86. We have,

$$\begin{aligned}
 (1+x)^n &= {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \\
 &\quad \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \quad \dots(i)
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } (1+x)^n &= {}^nC_n x^n + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \\
 &\quad \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \quad \dots(ii)
 \end{aligned}$$

Multiplying Eqs (i) and (ii), we get

$$\begin{aligned}
 (1+x)^{2n} &= ({}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} \\
 &\quad + {}^nC_2 \cdot {}^nC_{n-2} + \dots + {}^nC_n \cdot {}^nC_0) x^n \\
 &\quad + (\dots)x + (\dots)x^2 + \dots
 \end{aligned}$$

Comparing the co-efficients of x^n from both the sides, we get

$$\begin{aligned}
 &({}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} + {}^nC_2 \cdot {}^nC_{n-2} \\
 &\quad + \dots + {}^nC_n \cdot {}^nC_0) = {}^{2n}C_n
 \end{aligned}$$

87. We have,

$$(1+x)^n = {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \dots (i)$$

$$\text{Also, } (x+1)^n = {}^nC_0 \cdot x^n + {}^nC_1 \cdot x^{n-1} + {}^nC_2 \cdot x^{n-2} + \dots + {}^nC_r \cdot x^{n-r} + {}^nC_{r+1} \cdot x^{n-r-1} + \dots + {}^nC_n \dots (ii)$$

Multiplying Eqs (i) and (ii), we get

$$(1+x)^{2n} = ({}^nC_r \cdot {}^nC_n + {}^nC_{r-1} \cdot {}^nC_{n-1} + {}^nC_{r-2} \cdot {}^nC_{n-2} + \dots + {}^nC_0 \cdot {}^nC_n) x^r + (\dots)x + (\dots)x^2$$

Comparing the co-efficients of x^{n+r} from both the sides we get,

$$({}^nC_r \cdot {}^nC_n + {}^nC_{r-1} \cdot {}^nC_{n-1} + \dots + {}^nC_0 \cdot {}^nC_n) = {}^{2n}C_{n+r} = {}^{2n}C_{n-r}$$

88. We have,

$$(1+x)^n = {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \dots (i)$$

$$\text{Also, } (x+1)^n = {}^nC_0 \cdot x^n + {}^nC_1 \cdot x^{n-1} + {}^nC_2 \cdot x^{n-2} + \dots + {}^nC_r \cdot x^{n-r} + {}^nC_{r+1} \cdot x^{n-r-1} + \dots + {}^nC_n \dots (ii)$$

Multiplying Eqs (i) and (ii), we get

$$(1+x)^{2n} = [({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2] x^n + (\dots)x + (\dots)x^2 + \dots$$

Comparing the co-efficients of x^n from both the sides, we get

$$(({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2) = {}^{2n}C_n$$

89. We have,

$$(1+x)^n = {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + \dots + {}^nC_r \cdot x^r + {}^nC_{r+1} \cdot x^{r+1} + \dots + {}^nC_n \cdot x^n \dots (i)$$

$$\text{Also, } (x+1)^n = {}^nC_0 \cdot x^n + {}^nC_1 \cdot x^{n-1} + {}^nC_2 \cdot x^{n-2} + \dots + {}^nC_r \cdot x^{n-r} + {}^nC_{r+1} \cdot x^{n-r-1} + \dots + {}^nC_n \dots (ii)$$

Multiplying Eqs (i) and (ii), we get

$$(1+x)^{2n} = ({}^nC_0 \cdot {}^nC_1 + {}^nC_1 \cdot {}^nC_2 + {}^nC_2 \cdot {}^nC_3 + \dots + {}^nC_{n-1} \cdot {}^nC_n) x^{n+1} + (\dots)x + (\dots)x^2 + \dots$$

Comparing the co-efficients of x^{n+1} from both the sides, we get

$$({}^nC_0 \cdot {}^nC_1 + {}^nC_1 \cdot {}^nC_2 + {}^nC_2 \cdot {}^nC_3 + \dots + {}^nC_{n-1} \cdot {}^nC_n) = {}^{2n}C_{n+1} = {}^{2n}C_{n-1}$$

90. The co-efficient of $a^3b^4c^2d$ in the expansion of $(a+b+c+d)^{15}$ is

$$= \frac{15!}{5!4!2!1!1!}$$

91. Let $(ab)^x(bc)^y(ca)^z = a^3b^4c^7$

$$\Rightarrow a^{x+z} \cdot b^{x+y} \cdot c^{y+z} = a^3b^4c^7$$

$$\Rightarrow x+z=3, x+y=4, y+z=7$$

$$x=0, y=4, z=3$$

Thus,

the co-efficient of $a^3b^4c^7$ in the expansion of

$$(ab+bc+ca)^8$$

= the co-efficient of $(ab)^{10}(bc)^4(ca)^3$ in the expansion of $(ab+bc+ca)^8$

$$= \frac{8!}{0!4!3!} = 280.$$

92. Here $n=53$ and $m=5$.

So, the quotient is 10 and the remainder is 3.

Thus, the greatest co-efficient

$$= \frac{53!}{(10!)^{5-3} \times ((10+1)!)^3} = \frac{53!}{(10!)^2 \times (11!)^3}$$

93. The number of terms in the expansion of

$$(i) (a+b+c)^n \text{ is } {}^{n+3-1}C_{3-1} = {}^{n+2}C_2$$

$$(ii) (a+b+c+d)^n \text{ is } {}^{n+4-1}C_{4-1} = {}^{n+3}C_3$$

$$(iii) (a+b+c+d+e)^n \text{ is } {}^{n+5-1}C_{5-1} = {}^{n+4}C_4$$

94. The co-efficient of x^7 in the expansion of

$$(1+3x-2x^3)^{10}$$

$$= \sum \left(\frac{10!}{n_1!n_2!n_3!} \right) \times (1)^{n_1} (3)^{n_2} (-2)^{n_3}$$

where $n_1 + n_2 + n_3 = 10, n_2 + 3n_3 = 7$.

The possible values of n_1, n_2, n_3 are shown in tabular form.

n_1	n_2	n_3
3	7	0
5	4	1
7	1	2

Hence,

the co-efficient of x^7

$$= \frac{10!}{3!7!0!} \times (1)^3 (3)^7 (-2)^0 + \frac{10!}{5!4!1!} \times (1)^5 (3)^4 (-2)^1 + \frac{10!}{7!1!2!} \times (1)^7 (3)^1 (-2)^2 = 262440 - 204120 + 4320 = 62640$$

95. We have,

$$\frac{1}{1-3x}$$

$$= 1 + 3x + (3x)^2 + (3x)^3 + \dots + (3x)^n + \dots$$

Co-efficient of x^n in the expansion of $\frac{1}{1-3x} = 3^n$.

96. We have,

$$\begin{aligned}\frac{1}{1+4x} &= (1+4x)^{-1} \\ &= 1 - 4x + (-4x)^2 + \dots + (-4x)^n + \dots\end{aligned}$$

Co-efficient of x^n in the expansion of $\frac{1}{1+4x} = (-1)^n 4^n$.

97. We have,

$$\begin{aligned}\frac{1}{1-9x+20x^2} &= \frac{1}{1-(4+5)x+20x^2} \\ &= \frac{1}{(1-4x)(1-5x)} \\ &= \frac{5}{(1-5x)} - \frac{4}{(1-4x)}\end{aligned}$$

Co-efficient of x^n in the expansion of

$$\begin{aligned}\frac{1}{1-9x+20x^2} &= 5 \cdot 5^n - 4 \cdot 4^n \\ &= 5^{n+1} - 4^{n+1}\end{aligned}$$

98. We have,

$$\begin{aligned}\frac{1}{1-(a+b)x+abx^2} &= \frac{1}{(1-ax)(1-bx)} \\ &= \frac{1}{a-b} \left(\frac{a}{1-ax} - \frac{b}{1-bx} \right) \\ &= \frac{1}{a-b} [a(1-ax)^{-1} - b(1-bx)^{-1}]\end{aligned}$$

$$\begin{aligned}\therefore \text{Co-efficient of } x^n &= \frac{1}{a-b} (a \cdot a^n - b \cdot b^n) \\ &= \frac{a^{n+1} - b^{n+1}}{a-b}\end{aligned}$$

99. We have,

$$\begin{aligned}\left(\frac{1+x}{1-x} \right) &= (1+x)(1-x)^{-1} \\ &= (1+x)(1+x+x^2+x^3+\dots+x^{n-1}+x^n+\dots)\end{aligned}$$

$\therefore$ Co-efficient of $x^n = 1 + 1 = 2$

100. We have,

$$\begin{aligned}\left(\frac{1+x}{1-x} \right)^2 &= (1+x)^2(1-x)^{-2} \\ &= (1+2x+x^2)[1+2x+3x^2+4x^3+\dots \\ &\quad + (n-1)x^{n-2}+nx^{n-1}+(n+1)x^n+\dots]\end{aligned}$$

$\therefore$ Co-efficient of $x^n = n-1+2n+n+1 = 4n$

101. As we know that the general term in the expansion of $(1-x)^{-n}$ is

$$t_{r+1} = \frac{n(n+1)(n+2)\dots(n+r-1)}{r!} \times x^r$$

Thus, the co-efficient of x^n

$$\begin{aligned}&= \frac{\frac{1}{2}\left(\frac{1}{2}+1\right)\left(\frac{1}{2}+2\right)\dots\left(\frac{1}{2}+n-1\right)}{n!} \times (2)^n \\ &= \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2^n \times n!} \times (2)^n \\ &= \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!}\end{aligned}$$

102. Let t_{r+1} th term be the negative term.

We have

$$t_{r+1} = \frac{n(n-1)(n-2)\dots(n-(r-1))}{r!} \times \left(\frac{3}{4}x\right)^r$$

Thus, t_{r+1} is negative when $(n-r+1) < 0$

$$\Rightarrow \frac{13}{3} - r + 1 < 0$$

$$\Rightarrow r > \frac{16}{3} = 5\frac{1}{3}$$

$$\Rightarrow r = 6$$

103. We have,

$$\begin{aligned}y &= x - x^2 + x^3 - x^4 + \dots \\ \Rightarrow -y &= x - x^2 + x^3 - x^4 + \dots \\ \Rightarrow 1-y &= 1-x+x^2-x^3+x^4-\dots \\ \Rightarrow 1-y &= (1+x)^{-1} \\ \Rightarrow 1-y &= \frac{1}{(1+x)} \\ \Rightarrow y &= 1 - \frac{1}{1+x} = \frac{x}{1+x}\end{aligned}$$

104. We have,

$$\begin{aligned}y &= 2x + 3x^2 + 4x^3 + \dots \\ \Rightarrow 1+y &= 1 + 2x + 3x^2 + 4x^3 + \dots \\ \Rightarrow 1+y &= (1-x)^{-2} = \frac{1}{(1-x)^2} \\ \Rightarrow (1-x)^2 &= \frac{1}{1+y} \\ \Rightarrow 1-x &= \sqrt{\frac{1}{1+y}} \\ \Rightarrow x &= 1 - \sqrt{\frac{1}{1+y}}\end{aligned}$$

105. We have,

$$\begin{aligned}(1+x)^n &= 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots\end{aligned}$$

Comparing the co-efficient of x and x^2 , we get

$$\begin{aligned} nx &= \frac{1}{3}, \quad \frac{n(n-1)}{2} x^2 = \frac{1}{6} \\ \Rightarrow nx &= \frac{1}{3}, \quad \frac{nx(nx-x)}{2} = \frac{1}{6} \\ \Rightarrow \frac{nx(nx-x)}{2} &= \frac{1}{6} \\ \Rightarrow nx(nx-x) &= \frac{1}{3} \\ \Rightarrow \frac{1}{3} \left(\frac{1}{3} - x \right) &= \frac{1}{6} \\ \Rightarrow \left| \frac{1}{3} - x \right| &= 1 \\ \Rightarrow x &= \frac{1}{3} - 1 = -\frac{2}{3} \end{aligned}$$

Put $x = -\frac{2}{3}$, we get,

$$\begin{aligned} n \times -\frac{2}{3} &= \frac{1}{3} \\ \Rightarrow n &= -\frac{1}{2} \end{aligned}$$

Therefore,

$$\begin{aligned} 1 + \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots &= (1+x)^n \\ &= \left(1 - \frac{2}{3}\right)^{-1/2} \\ &= \left(\frac{1}{3}\right)^{-1/2} \\ &= \sqrt{3} \end{aligned}$$

106. We have,

$$\begin{aligned} &\frac{(1+2x)^{1/2} (16+3x)^{1/4}}{(1-x)^2} \\ &= \frac{(1+2x)^{1/2} \left[16 \left(1 + \frac{3}{16}x\right)\right]^{1/4}}{(1-x)^2} \\ &= \frac{2(1+2x)^{1/2} \left(1 + \frac{3}{16}x\right)^{1/4}}{(1-x)^2} \\ &= \frac{2 \left(1 + \frac{1}{2} \cdot 2x + \dots\right) \left(1 + \frac{1}{4} \cdot \frac{3}{16}x + \dots\right)}{(1-x)^2} \\ &= \frac{2(1+x) \left(1 + \frac{3}{64}x\right)}{(1-x)^2} \\ &= 2 \left(1 + x + \frac{3}{64}x\right) \times (1-x)^{-2} \\ &= 2 \left(1 + x + \frac{3}{64}x\right) \times (1+2x+\dots) \end{aligned}$$

$$\begin{aligned} &= 2 \left(1 + \frac{67}{64}x\right) \times (1+2x+\dots) \\ &= 2 \left(1 + \frac{67}{64}x + 2x\right) \\ &= 2 \left(1 + \frac{67}{64} \cdot \frac{1}{2} + 2 \cdot \frac{1}{2}\right) \\ &= 2 \left(2 + \frac{67}{128}\right) \\ &= \left|\frac{256+67}{64}\right| = \frac{323}{67} \end{aligned}$$

107. We have,

$$\begin{aligned} &\frac{(1+2x)^{1/2} + (16+3x)^{1/4}}{(1-2x)^{1/4}} \\ &= \frac{(1+x+\dots) + 2 \left(1 + \frac{3}{16}x\right)^{1/4}}{(1-2x)^{1/4}} \\ &= \frac{(1+x+\dots) + 2 \left(1 + \frac{1}{4} \cdot \frac{3}{16}x + \dots\right)}{(1-2x)^{1/4}} \\ &= \frac{(1+x) + \left(2 + \frac{3}{32}x\right)}{(1-2x)^{1/4}} \\ &= \left(3 + \frac{35}{32}x\right) \times (1-2x)^{-1/4} \\ &= \left(3 + \frac{35}{32}x\right) \times \left(1 + \frac{1}{4} \cdot 2x + \dots\right) \\ &= \left(3 + \frac{35}{32}x\right) \times \left(1 + \frac{x}{2}\right) \\ &= 3 + \frac{35x}{32} + \frac{3x}{2} \\ &= \left|3 + \frac{83x}{32}\right| \end{aligned}$$

Thus, $a = 3$ and $b = \frac{83}{32}$

108. We have,

$$\begin{aligned} &\sqrt{\frac{a}{a+x}} + \sqrt{\frac{a}{a-x}} \\ &= \sqrt{\frac{1}{\left(1+\frac{x}{a}\right)}} + \sqrt{\frac{1}{\left(1-\frac{x}{a}\right)}} \\ &= \left(1+\frac{x}{a}\right)^{-1/2} + \left(1-\frac{x}{a}\right)^{-1/2} \\ &= \left(1 - \frac{x}{2a} + \frac{3x^2}{8a^2} + \dots\right) + \left(1 + \frac{x}{a} + \frac{3x^2}{8a^2} + \dots\right) \\ &= 2 + \frac{3x^2}{4a^2} + \dots \end{aligned}$$

109. Let $p = q + h$

$$\Rightarrow \frac{p}{q} = 1 + \frac{h}{q}$$

We have,

$$\begin{aligned} & \frac{5p+4q}{4p+5q} \\ &= \frac{5\left(\frac{p}{q}\right) + 4}{4\left(\frac{p}{q}\right) + 5} \\ &= \frac{5\left(1 + \frac{h}{q}\right) + 4}{4\left(1 + \frac{h}{q}\right) + 5} \\ &= \frac{\left(1 + \frac{5h}{9q}\right)}{\left(1 + \frac{4h}{9q}\right)} \\ &= \left(1 + \frac{5h}{9q}\right) \times \left(1 + \frac{4h}{9q}\right)^{-1} \\ &= \left(1 + \frac{5h}{9q}\right) \times \left(1 - \frac{4h}{9q}\right) \\ &= \left|1 + \frac{5h}{9q} - \frac{4h}{9q}\right| \\ &= \left|1 + \frac{h}{9q}\right| = \left(1 + \frac{h}{q}\right)^{1/9} \\ &= \left(\frac{p}{q}\right)^{1/9} \end{aligned}$$

110. Let $x = 1 + h$

We have,

$$\begin{aligned} & \frac{mx^m - nx^n}{m-n} \\ &= \frac{m(1+h)^m - n(1+h)^n}{m-n} \\ &= \frac{m(1+mh) - n(1+nh)}{m-n} \\ &= \frac{(m-n) - (m^2 - n^2)h}{m-n} \\ &= 1 - (m+n)h \\ &= (1-h)^{(m+n)} \\ &= x^{(m+n)} \end{aligned}$$

111. We have,

$$e^{5x} = 1 + 5x + \frac{(5x)^2}{2!} + \frac{(5x)^3}{3!} + \frac{(5x)^4}{4!} + \dots + \frac{(5x)^n}{n!} + \dots$$

Thus,

$$\text{the co-efficient of } x^n \text{ in } e^{5x} = \frac{5^n}{n!}$$

112. We have,

$$\begin{aligned} & e^{5x+4} \\ &= e^4 \cdot e^{5x} \\ &= e^4 \left(1 + \frac{5x}{1} + \frac{(5x)^2}{2!} + \frac{(5x)^3}{3!} + \dots + \frac{(5x)^n}{n!} + \dots \right) \end{aligned}$$

Thus,

$$\text{the co-efficient of } x^n \text{ in } e^{5x+4} = e^4 \times \left(\frac{5^n}{n!} \right)$$

113. We have,

$$\begin{aligned} & \left(\frac{1+3x+2x^2}{e^x} \right) \\ &= (1+3x+2x^2) \times e^{-x} \\ &= (1+3x+2x^2) \times \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^n}{n!} + \dots \right) \end{aligned}$$

Thus,

the co-efficient of x^n

$$= \frac{(-1)^n}{n!} + \frac{3 \cdot (-1)^{n-1}}{(n-1)!} + \frac{2 \cdot (-1)^{n-2}}{(n-2)!}$$

114. Given,

$$\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \dots + B_nx^n$$

Now,

$$\begin{aligned} & e^x \times (1-x)^{-1} \\ &= \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} + \dots \right) \\ & \quad \times (1 + x + x^2 + x^3 + \dots + x^n + \dots) \\ &= \left(\frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{1!} + 1 \right) x^n \\ & \quad + \left(\frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \frac{1}{(n-3)!} + \dots + \frac{1}{1!} + 1 \right) x^{n-1} \end{aligned}$$

Thus, comparing the co-efficients of x^{n-1} and x^n , we get

$$B_n = \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{1!} + 1$$

and

$$B_{n-1} = \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \frac{1}{(n-3)!} + \dots + \frac{1}{1!} + 1$$

$$\text{Therefore, } B_n - B_{n-1} = \frac{1}{n!}.$$

115 We have,

$$1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots = \frac{1}{2}(e + e^{-1})$$

$$\text{and } 1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots = \frac{1}{2}(e - e^{-1})$$

$$\begin{aligned} \text{Now, } & \frac{1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots}{1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots} \\ &= \frac{\frac{1}{2}(e + e^{-1})}{\frac{1}{2}(e - e^{-1})} \\ &= \frac{e^2 + 1}{e^2 - 1} \end{aligned}$$

116. We have,

$$\begin{aligned} x &= \left(1 + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots\right)^2 \\ &\quad - \left(1 + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots\right)^2 \\ &= \frac{1}{4}(e + e^{-1})^2 - \frac{1}{4}(e - e^{-1})^2 \\ &= \frac{1}{4}\left(e^2 + \frac{1}{e^2} + 2\right) - \frac{1}{4}\left(e^2 + \frac{1}{e^2} - 2\right) \\ &= \frac{1}{4}\left(e^2 + \frac{1}{e^2} + 2 - e^2 - \frac{1}{e^2} + 2\right) \\ \Rightarrow x &= \frac{1}{4}(4) = 1 \end{aligned}$$

Therefore,

$$\begin{aligned} (x+1)^{x+1} + 10 &= (2)^2 + 10 \\ &= 4 + 10 \\ &= 14 \end{aligned}$$

117. We have,

$$\begin{aligned} a &= \sum_{n=1}^{\infty} \left(\frac{2n}{(2n-1)!} \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{(2n-1)+1}{(2n-1)!} \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{(2n-2)!} + \frac{1}{(2n-1)!} \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{(2n-2)!} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{(2n-1)!} \right) \\ &= \frac{1}{2}(e + e^{-1}) - \frac{1}{2}(e - e^{-1}) \\ &= e \end{aligned}$$

$$\begin{aligned} \text{Also, } b &= \sum_{n=1}^{\infty} \left(\frac{2n}{(2n+1)!} \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{(2n+1)-1}{(2n+1)!} \right) \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{(2n)!} - \frac{1}{(2n+1)!} \right) \end{aligned}$$

$$\begin{aligned} &= \sum_{n=1}^{\infty} \left(\frac{1}{(2n)!} \right) - \sum_{n=1}^{\infty} \left(\frac{1}{(2n+1)!} \right) \\ &= \frac{1}{2}(e + e^{-1} - 1) - \frac{1}{2}(e - e^{-1} - \frac{1}{2}) \\ &= \frac{1}{e} \end{aligned}$$

Therefore, $ab = 1$.

$$118. \text{ Let } t_n = \frac{1+2+3+\dots+n}{n!}$$

$$\begin{aligned} &= \frac{n(n+1)}{2 \times n!} \\ &= \frac{1}{2} \frac{(n+1)}{(n-1)!} \\ &= \frac{1}{2} \frac{[(n-1)+2]}{(n-1)!} \\ &= \frac{1}{2} \left(\frac{1}{(n-2)!} + \frac{1}{(n-1)!} \right) \end{aligned}$$

Therefore, $S = \sum_{n=1}^{\infty} t_n$

$$\begin{aligned} &= \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{(n-2)!} + \frac{1}{(n-1)!} \right) \\ &= \frac{1}{2} \left[\sum_{n=1}^{\infty} \left(\frac{1}{(n-2)!} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{(n-1)!} \right) \right] \\ &= \frac{1}{2}(e + e) \\ &= e \end{aligned}$$

119. We have,

$$\begin{aligned} & \frac{C(n, 4)}{P(n, n)} \\ &= \frac{{}^nC_4}{n!} \\ &= \frac{(n)!}{4! \times (n-4)!} \times \frac{1}{n!} \\ &= \frac{1}{4! \times (n-4)!} \\ &= \frac{1}{24} \times \frac{1}{(n-4)!} \end{aligned}$$

Therefore,

$$\begin{aligned} \sum_{n=4}^{\infty} \left(\frac{C(n, 4)}{P(n, n)} \right) &= \sum_{n=4}^{\infty} \frac{1}{24} \times \frac{1}{(n-4)!} \\ &= \frac{1}{24} \left(\sum_{n=4}^{\infty} \frac{1}{(n-4)!} \right) \\ &= \frac{1}{24} \times e \\ &= \frac{e}{24} \end{aligned}$$

120. We have,

$$a = \sum_{n=0}^{\infty} \left(\frac{x^{3n}}{(3n)!} \right) \\ = \left(1 + \frac{x^3}{3!} + \frac{x^6}{6!} + \frac{x^9}{9!} + \dots \right)$$

Also,

$$b = \sum_{n=0}^{\infty} \left(\frac{x^{3n-2}}{(3n-2)!} \right) \\ = \left(\frac{x}{1} + \frac{x^4}{4!} + \frac{x^7}{7!} + \frac{x^{10}}{10!} + \dots \right)$$

Further, $c = \sum_{n=0}^{\infty} \left(\frac{x^{3n-1}}{(3n-1)!} \right)$

$$= \left(\frac{x^2}{2!} + \frac{x^5}{5!} + \frac{x^8}{8!} + \frac{x^{11}}{11!} + \dots \right)$$

Now,

$$a + b + c = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots = e^x$$

$$a + b\omega + c\omega^2 = e^{a\omega}$$

and $a + b\omega^2 + c\omega = e^{a\omega^2}$

Therefore, $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$$

$$= e^x \cdot e^{a\omega} \cdot e^{a\omega^2}$$

$$= e^{x(1+\omega+\omega^2)}$$

$$= e^{x \cdot 0} = 1$$

121. As we know that,

$$2 < e < 3$$

$$\Rightarrow \log_e 2 < \log_e e < \log_e 3$$

$$\Rightarrow \log_e 2 < 1 < \log_e 3$$

Hence, the result.

122. As we know that,

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Put $x = \frac{1}{4}$, we get

$$\frac{1}{4} - \frac{1}{2} \left(\frac{1}{4} \right)^2 + \frac{1}{3} \left(\frac{1}{4} \right)^3 - \frac{1}{4} \left(\frac{1}{4} \right)^4 + \dots \\ = \log_e \left(1 + \frac{1}{4} \right) = \log_e \left(\frac{5}{4} \right)$$

123. As we know that,

$$\log \left(\frac{1+x}{1-x} \right) = 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots \right)$$

Replacing x by $\frac{1}{2x+1}$, we get

$$2 \left(\frac{1}{(2x+1)} + \frac{1}{3(2x+1)^3} + \frac{1}{5(2x+1)^5} + \dots \right) \\ = \log \left(\frac{1 + \frac{1}{2x+1}}{1 - \frac{1}{2x+1}} \right)$$

$$= \log \left(\frac{\frac{2x+1+1}{2x+1}}{\frac{2x+1-1}{2x+1}} \right) \\ = \log \left(\frac{2x+2}{2x} \right) \\ = \log \left(\frac{x+1}{x} \right)$$

124. We have,

$$\log_e(1+6x+8x^2) \\ = \log_e(1+4x)(1+2x) \\ = \log_e(1+4x) + \log_e(1+2x) \\ = 4x - \frac{(4x)^2}{2} + \frac{(4x)^3}{3} - \frac{(4x)^4}{4} \\ + \dots + \frac{(-1)^{n+1}(4x)^n}{n} + \dots \\ + 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \frac{(2x)^4}{4} \\ + \dots + \frac{(-1)^{n+1}(2x)^n}{n} + \dots$$

Thus,

the co-efficient of x^n

$$= \frac{(-1)^{n+1}}{n} (4^n + 2^n)$$

125. Given,

$$y = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$\Rightarrow -y = - \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots \right)$$

$$\Rightarrow -y = \log_e(1-x)$$

$$\Rightarrow (1-x) = e^{-y}$$

$$\Rightarrow x = 1 - e^{-y}$$

$$= 1 - \left(1 - y + \frac{y^2}{2!} - \frac{y^3}{3!} + \frac{y^4}{4!} - \frac{y^5}{5!} + \dots \right) \\ = \left(y - \frac{y^2}{2!} + \frac{y^3}{3!} - \frac{y^4}{4!} + \frac{y^5}{5!} - \dots \right)$$

126. We have,

$$\log_e(1+x+x^2+x^3+x^4+\dots)$$

$$= \log_e \left(\frac{1}{1-x} \right)$$

$$= \log_e 1 - \log_e(1-x)$$

$$= -\log_e(1-x)$$

$$= \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \dots \right)$$

Hence, the result.

127. We have,

$$\begin{aligned}
 & \frac{1}{2 \cdot 3} + \frac{1}{4 \cdot 5} + \frac{1}{6 \cdot 7} + \dots \\
 &= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{6} - \frac{1}{7} \right) + \dots \\
 &= - \left(-\frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \dots \right) \\
 &= 1 - \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \dots \right) \\
 &= 1 - \log_e 2 \\
 &= \log_e e - \log_e 2 \\
 &= \log_e \left(\frac{e}{2} \right)
 \end{aligned}$$

128. Let $m = \frac{1}{(2n-1)2n(2n+1)}$

$$\begin{aligned}
 &= \frac{1}{2} \left(\frac{1}{(2n-1)2n} - \frac{1}{(2n-1)2n(2n+1)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{(2n-1)2n} - \frac{1}{(2n-1)2n} + \frac{1}{(2n-1)2n(2n+1)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{(2n-1)2n} - \frac{1}{(2n-1)2n} + \frac{1}{(2n-1)2n(2n+1)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{(2n-1)2n} - \frac{1}{(2n-1)2n} + \frac{1}{(2n-1)2n(2n+1)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{(2n-1)2n} - \frac{1}{(2n-1)2n} + \frac{1}{(2n-1)2n(2n+1)} \right)
 \end{aligned}$$

Therefore,
the sum

$$\begin{aligned}
 &= \frac{1}{2} \left[\left(1 + \frac{1}{3} + \frac{1}{5} + \dots \right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \right) \right. \\
 &\quad \left. + G + \frac{1}{4} + \dots \right] \\
 &= \frac{1}{2} \left(-\frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots \right) \\
 &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots \right) - \frac{1}{2} \\
 &= \frac{1}{2} \log_e 2 - \frac{1}{2} \\
 &= \frac{1}{2} (\log_e 2 - 1) \\
 &= \frac{1}{2} (\log_e \left(\frac{e}{2} \right))
 \end{aligned}$$

129. We have

$$\begin{aligned}
 & 1 + \left(\frac{1}{2} + \frac{1}{3} \right) \frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5} \right) \frac{1}{4^2} + \left(\frac{1}{6} + \frac{1}{7} \right) \frac{1}{4^3} + \dots \\
 &= \left(\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot \left(\frac{1}{4} \right)^2 + \left(\frac{1}{6} \right) \cdot \left(\frac{1}{4} \right)^3 + \dots \right) \\
 &\quad + \left(1 + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{5} \cdot \left(\frac{1}{4} \right)^2 + \left(\frac{1}{7} \right) \cdot \left(\frac{1}{4} \right)^3 + \dots \right) \\
 &= -\frac{\log \left(1 - \frac{1}{16} \right)}{2} + \log \left(\frac{\left(1 + \frac{1}{4} \right)}{\left(1 - \frac{1}{4} \right)} \right) \\
 &= -\frac{\log \left(\frac{15}{16} \right)}{2} + \frac{1}{2} \log \left(\frac{5}{3} \right) \\
 &= -\frac{1}{2} \left[\log \left(\frac{15}{16} \right) - \log \left(\frac{5}{3} \right) \right] \\
 &= -\frac{1}{2} \log \left(\frac{15}{16} \times \frac{3}{5} \right) \\
 &= -\frac{1}{2} \log \left(\frac{9}{16} \right) \\
 &= -\frac{1}{2} \log \left(\frac{3}{4} \right)^2 \\
 &= -\log \left(\frac{3}{4} \right) \\
 &= \log \left(\frac{4}{3} \right)
 \end{aligned}$$

130. Let $\left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) = x$

Then,
the sum

$$\begin{aligned}
 &= 1 + \frac{x}{2} + \frac{x^2}{6} + \frac{x^3}{12} + \frac{x^4}{20} + \dots \\
 &= 1 + \frac{x}{1 \cdot 2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{3 \cdot 4} + \frac{x^4}{4 \cdot 5} + \dots \\
 &= [1 + (1 - \frac{1}{2})x + (\frac{1}{2} - \frac{1}{3})x^2 \\
 &\quad + G - \frac{1}{3}x^3 + (\frac{1}{3} - \frac{1}{4})x^4 + \dots] \\
 &= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \right) \\
 &\quad - \left(\frac{x}{2} + \frac{x^2}{3} + \frac{x^3}{4} + \dots \right)
 \end{aligned}$$